# Applied Study

Supplementary material for the CountBCF manuscript

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## 1 Applied Study

The simulation experiments establish the operating characteristics of CountBCF under controlled data-generating mechanisms. We now assess it in a setting where the outcome, exposure, and scientific benchmarks are all real. We analyse solid-cancer incidence in the Radiation Effects Research Foundation (RERF) Life Span Study (LSS) of survivors of the Hiroshima and Nagasaki atomic bombings. The LSS is the principal source of quantitative evidence on radiation-related cancer risk in humans, with detailed follow-up and well-characterised dosimetry supporting successive incidence analyses [Thompson et al., 1994, Preston et al., 2007, Grant et al., 2017].

The application has two purposes. First, it is a convergence check: on an outcome for which carefully constructed excess-relative-risk models are available, do the marginal effects recovered by CountBCF agree with the corresponding parametric analysis? Second, it is a substantive demonstration: can a nonparametric treatment-effect surface recover established variation in radiation risk without the analyst pre-specifying interactions in sex, age at exposure, or attained age? The answer to both questions is affirmative for the well-powered all-solid-cancer contrast. CountBCF also flags one candidate modifier that the canonical model class structurally cannot express, a city difference in the low-dose contrast. Following that flag with the standard epidemiological tools localises it to the composition of the zero-dose reference group rather than to radiation, which is the intended use of the method: the forest identifies where to look, and the domain analysis then says what is there.

This document is the supplementary version of the applied study and reports the complete set of tables, figures, sensitivity fits, and diagnostics. A condensed version appears in the main text. All analyses use the complete 1958–1998 person-year table; results are produced by the pipeline in `Applied_Study/code/`, with tables written to `Applied_Study/analysis/tables/` and figures to `Applied_Study/analysis/plots/`. All random seeds are fixed, and `run_all.sh` records `sessionInfo()` and the full package version list alongside the outputs, so a clean clone reproduces both the pipeline and the numbers.

### 1.1 Data validation against the published analysis

Before any modelling, the pipeline verifies that our handling of the public file reproduces the published totals. Table 1 compares our analysis frame with Table 4 of Preston et al. [2007]. Every

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case count and every head count matches exactly, and person-years agree to a maximum relative difference of $2.6 \times 10^{-4}$, which is attributable to rounding in the published table. Because the data-handling step is verified to the case, any disagreement in the results that follow is a modelling difference rather than a data-handling difference.

Table 1: Validation of the analysis frame against Table 4 of Preston et al. [2007]. Case and person counts match exactly; the person-year differences reflect rounding in the published table.

| Stratum | Cases | Published | Person-years | Published | People | Published |
|---|---|---|---|---|---|---|
| Not in city | 3,994 | 3,994 | 680,744 | 680,744 | 25,247 | 25,247 |
| $<0.005$ Gy | 5,603 | 5,603 | 918,200 | 918,199 | 35,545 | 35,545 |
| 0.005–0.1 | 4,406 | 4,406 | 729,604 | 729,603 | 27,789 | 27,789 |
| 0.1–0.2 | 968 | 968 | 145,925 | 145,925 | 5,527 | 5,527 |
| 0.2–0.5 | 1,144 | 1,144 | 153,886 | 153,846 | 5,935 | 5,935 |
| 0.5–1 | 688 | 688 | 81,251 | 81,251 | 3,173 | 3,173 |
| 1–2 | 460 | 460 | 41,413 | 41,412 | 1,647 | 1,647 |
| 2–4 | 185 | 185 | 13,712 | 13,711 | 564 | 564 |
| Male | 7,969 | 7,969 | 1,040,281 | 1,040,278 | 42,902 | 42,902 |
| Female | 9,479 | 9,479 | 1,724,454 | 1,724,452 | 62,525 | 62,525 |
| Hiroshima | 12,931 | 12,931 | 1,967,602 | 1,967,601 | 73,430 | 73,430 |
| Nagasaki | 4,517 | 4,517 | 797,133 | 797,129 | 31,997 | 31,997 |
| Total | 17,448 | 17,448 | 2,764,735 | 2,764,730 | 105,427 | 105,427 |

The composition of the all-solid endpoint also required verification, because it determines which site-specific mechanisms can drive the aggregate result. The all-solid total decomposes cell by cell, with zero discrepancies, as the sum of oral cavity, digestive, respiratory, skin/bone/connective tissue, breast, female genital, male genital, urinary, renal pelvis and ureter, central nervous system, thyroid, other solid, and thymus. Two codebook features make this non-obvious: the urinary group (661 known-dose cases) omits renal pelvis and ureter (80 cases) even though the documentation places the latter under the urinary system, and thymus (12 cases) sits outside the respiratory group. Restoring both recovers 17,448 exactly. Group by group against Table 2 of Preston et al. [2007], oral cavity (277), digestive (10,052), respiratory (2,001), skin, bone and connective tissue (400), breast (1,082), female genital (1,457), male genital (420), central nervous system (281), thyroid (471), stomach (4,730), lung (1,759), colon (1,516) and liver (1,494) all match the published counts to the case. Within the skin, bone and connective tissue group, non-melanoma skin cancer contributes 330 cases and melanoma 17, again exactly reproducing the published figures. Non-melanoma skin cancer is therefore inside the all-solid aggregate, consistent with the published description of 17,448 first primary cancers including non-melanoma skin cancer. This matters because non-melanoma skin is one of the few sites whose ERR varies significantly with age at exposure in the published heterogeneity tests, making it a candidate driver of the aggregate's age-at-exposure shape. Section 1.9 tests that candidate directly.

## 1.2 Data, treatment design, and estimands

The source is the public 1958–1998 LSS solid-cancer incidence person-year table. Each row is a stratum defined by city, sex, age at exposure, attained age, and follow-up period; it records cancer cases and person-years at risk. The analytic endpoint is all first primary solid cancers. Its 17,448 cases among 105,427 cohort members reproduce the published totals in Preston et al. [2007], as documented in Section 1.1.

Unknown-dose records are excluded: 1,236 cells carrying 174,631 person-years and 1,197 cases. The primary analysis is restricted to in-city survivors and uses the standard DS02 weighted colon dose, gamma dose plus ten times neutron dose, expressed in Gy. We contrast a common reference group below 5 mGy against three exposure bands: low (5–< 100 mGy), medium (100–< 500 mGy), and high (500–< 2000 mGy). Survivors at or above 2 Gy are excluded from the primary contrasts and restored in a sensitivity fit. Each band is fitted as a binary CountBCF analysis against the same reference group. This categorical construction makes the causal contrast explicit and does not impose a dose-response curve. One point of nomenclature deserves mention: the dose columns are weighted, adjusted, truncated DS02 organ doses in Gy despite the codebook labelling them mGy, whereas the dose-category cutpoints are in mGy.

Survivors not in either city are excluded from the primary contrast because their probability of receiving an in-city bomb dose is zero, so including them in the control arm of a treated-versus-control contrast would violate the overlap condition that CountBCF relies on. This is not an objection to their use in a regression that merely borrows their follow-up to pin down the baseline shape, which is the conventional LSS treatment. We therefore include them in a sensitivity analysis with a not-in-city indicator in the prognostic forest, the CountBCF analogue of the standard baseline adjustment, and compare the two fits in Section 1.9. The published analysis provides the benchmark for this check, reporting that dose parameters are virtually unchanged when the not-in-city group is excluded and that including it without baseline adjustment inflates the standardised ERR by about 8%.

The prognostic forest receives city, sex, age at exposure, attained age, and the estimated propensity score; the moderating forest receives the same substantive covariates. Calendar year, birth cohort, and time since exposure are deterministic functions of the two age variables and are not entered separately. The relation is year = age − age at exposure + $t_0$, where $t_0$ is the date of the detonation in decimal years. Across all 26,806 cells the implied $t_0$ lies between 1945.594 and 1945.611, with a median of 1945.597 in Hiroshima and 1945.605 in Nagasaki, which are 6 and 9 August 1945. The relation is thus exact to within about six days and is a property of the calendar rather than of the coding. City is placed in the moderator set deliberately. The canonical risk models put city in the baseline only and never allow it to modify the ERR, so a credible city effect on the conditional effect would be a modifier that the canonical model cannot express. Ground distance from the hypocentre is omitted because it is part of the dose-assignment mechanism rather than a pre-exposure confounder; a preliminary analysis confirmed that conditioning on it collapses overlap and reverses the sign of the estimate. This exclusion is correct as a matter of identification, but it has a consequence that Sections 1.10, 1.11 and 1.15 return to: within a city the dose band is very nearly a distance ring, so anything distance-correlated and city-specific is unadjustable by construction in the primary specification. Section 1.10 therefore adds a separate family of fits in which the coarse $< 3/\geq 3$ km ring, which is a baseline-composition variable rather than the continuous assignment variable, does enter both the score and the prognostic forest. The propensity-score adjustment follows the regularization-induced-confounding strategy of Hahn et al. [2020], Rosenbaum and Rubin [1983].

With person-years as an offset, $\exp(\mu_{f,i})$ is the reference-arm rate in cell $i$ and $\tau_{f,i}$ is the log rate ratio. We report both an absolute and a relative conditional effect:

$$\mathrm{EAR}_i = \{\exp(\mu_{f,i} + \tau_{f,i}) - \exp(\mu_{f,i})\} \times 10^4, \qquad \mathrm{ERR}_i = \exp(\tau_{f,i}) - 1. \tag{1}$$

The first is an excess absolute rate in cases per $10^4$ person-years; the second is the ERR used in LSS risk modelling. The standardised ERR is the ratio of person-year-weighted counterfactual rates, $(\sum_i w_i r_{1,i}) \,/\, (\sum_i w_i r_{0,i}) - 1$, rather than an average of cell-specific rate ratios. The distinction is

quantitatively important, not pedantic: cells with a high ERR tend to be cells with a low baseline rate, so the average of ratios runs well above the ratio of averages. For high-dose all-solid cancer, averaging per-cell rate ratios would report a rate ratio of 1.84 against the standardised 1.49, creating a spurious disagreement with the parametric benchmark. Parametric g-computation returns the standardised quantity, so that is what we compare with the GLM rate ratio and with published case-weighted summaries, while the per-cell average is the conditional-effect average used in the subgroup separation tests [Hernán and Robins, 2006, Robins, 1986]. All summaries are computed on the posterior draws, so a subgroup contrast such as the difference between a conditional effect and the average effect carries a genuine credible interval rather than being a difference of two point estimates.

The causal interpretation is conditional on consistency, no interference, positivity, and conditional exchangeability given the available covariates. Dose resulted from location and shielding at the time of the detonations, not a behavioural treatment choice, but the study remains observational. In particular, the public table contains no individual smoking, dietary, or infection information. The estimates should therefore be read as causal under these assumptions and as adjusted associations otherwise.

### 1.3 Descriptive structure of the analysis table

Table 2: Descriptive summary of the all-solid-cancer analysis table. The reference and three treatment bands are in-city groups. Rates are crude and reported before covariate adjustment. The final column is the marginal variance-to-mean ratio of the cell counts.

| Dose band | Cells | Subjects | Person-years | Cases | Rate/$10^4$ PY | Mean dose (Gy) | Zero cells (%) | Var/mean |
|---|---|---|---|---|---|---|---|---|
| Not in city | 1,285 | 25,247 | 680,744 | 3,994 | 58.7 | 0 | 44.1 | 8.10 |
| Reference, 0–< 5 mGy | 2,568 | 35,545 | 918,200 | 5,603 | 61.0 | 0.00096 | 47.2 | 5.37 |
| Low, 5–< 100 mGy | 5,973 | 27,789 | 729,604 | 4,406 | 60.4 | 0.0308 | 68.6 | 3.42 |
| Medium, 100–< 500 mGy | 7,598 | 11,462 | 299,812 | 2,112 | 70.4 | 0.2354 | 80.8 | 1.65 |
| High, 500–< 2000 mGy | 5,681 | 4,820 | 122,664 | 1,148 | 93.6 | 0.9216 | 83.9 | 1.34 |

Table 2 illustrates why adjustment is necessary. The low-dose crude rate of 60.4 per $10^4$ person-years is slightly below the reference rate of 61.0, although the adjusted dose response rises across the medium and high bands. This is confounding rather than biology. Age at exposure, attained age, and sex are almost perfectly balanced across bands, with standardised differences between 0.002 and 0.047, whereas city is not: Hiroshima accounts for 58% of the reference group and 78% of the low-dose group, giving standardised differences between 0.19 and 0.56. City is therefore the one covariate on which the propensity model has real work to do, and it is why the adjusted analyses move the low-band estimate.

The cells are sparse but not, on inspection, genuinely overdispersed. Roughly 74% of the analysis cells are zero and the marginal variance-to-mean ratio runs from 1.3 to 5.4. Once a flexible baseline is conditioned on, consisting of sex-specific splines in log attained age and in age at exposure plus a city term, the extra-Poisson variation essentially disappears: the fitted negative-binomial size is $\theta \approx 100$, implying a variance-to-mean ratio of 1.006 at the mean fitted rate; the negative-binomial AIC is no better than the Poisson AIC; and the likelihood-ratio test for the negative binomial over the Poisson gives $p = 0.5$. The quasi-Poisson scale, at a Pearson $\chi^2/\mathrm{df}$ of 1.69, is inflated mostly by the sparse-cell behaviour of the Pearson statistic, since the deviance per degree of freedom is 0.59. The apparent overdispersion in this table is thus baseline heterogeneity rather than extra-Poisson noise, and a model flexible enough in the baseline absorbs it. We nevertheless run CountBCF with

the negative-binomial likelihood because it nests the Poisson and lets the data decide; its posterior for the dispersion, reported in Section 1.5, reaches the same conclusion from the same fit that produces the causal estimates. The many zero cells are sampling zeros in thin person-year strata rather than evidence of a not-at-risk subpopulation.

The crude data already display the canonical gradient, which is the pattern against which any model output must be read. Comparing the high band to the reference within age-at-exposure strata, the crude rate ratio runs 3.41, 1.90, 1.51, 1.35, and 1.41 across age-at-exposure groups 0–9, 10–19, 20–29, 30–39, and 40 or more for women, and 1.71, 1.72, 1.41, 1.12, and 1.20 for men.

## 1.4 Parametric anchors

Three parametric anchors are fitted on the identical in-city analytic set: a refit of the canonical excess-relative-risk and excess-absolute-risk models, a fully categorical dose response, and adjusted generalised linear models for the marginal contrasts.

### 1.4.1 Refitting the canonical risk model

The standard model is refitted by maximum Poisson likelihood on the same in-city analysis set, using a sex-specific spline baseline in log attained age and in age at exposure plus a city term, with

$$\mathrm{ERR}(d, e, a, s) = \beta_s \, d \, \exp\{\alpha_e (e - 30) + \alpha_a \log(a/70)\}\,, \tag{2}$$

and the additive analogue for the EAR, where $d$ is weighted colon dose, $e$ is age at exposure, $a$ is attained age, and $s$ is sex. Because this refit is the comparator against which the CountBCF surface is measured in Section 1.7, it matters that it reproduces the published fit. Table 3 shows that every one of the ten published parameters is recovered inside its published 90% confidence interval. The parametric yardstick is therefore the established risk model itself, rather than a simplified regression surrogate.

Table 3: Refit of the canonical ERR and EAR risk models on the in-city analytic set, compared with Table 10 of Preston et al. [2007]. Intervals are 90% confidence intervals. Every refitted parameter falls inside the published interval.

| Parameter | Refit (90% CI) | Published (90% CI) |
|---|---|---|
| ERR per Gy, male, attained age 70, exposed at 30 | 0.336 (0.245, 0.426) | 0.35 (0.28, 0.43) |
| ERR per Gy, female, attained age 70, exposed at 30 | 0.554 (0.438, 0.670) | 0.58 (0.43, 0.69) |
| ERR female-to-male ratio | 1.65 | 1.6 (1.31, 2.09) |
| ERR change per decade of age at exposure | −15.2% (−25.6%, −3.1%) | −17% (−25%, −7%) |
| ERR attained-age power | −1.87 (−2.45, −1.28) | −1.65 (−2.1, −1.2) |
| EAR per $10^4$ PY-Gy, male | 38.9 (25.8, 52.1) | 43 (33, 55) |
| EAR per $10^4$ PY-Gy, female | 57.8 (47.6, 68.1) | 60 (51, 69) |
| EAR female-to-male ratio | 1.49 | 1.4 (1.10, 1.79) |
| EAR change per decade of age at exposure | −23.5% | −24% (−32%, −16%) |
| EAR attained-age power | +2.25 (1.73, 2.77) | 2.38 (1.9, 2.8) |

Fitting one parameter per dose category on top of the same baseline gives 18 nonparametric dose points. A weighted linear fit through the origin gives an ERR of 0.61 per Gy with a standard error of 0.04 in the absence of effect modification. This is larger than the 0.47 standardised to attained age 70 and exposure at age 30, as expected, since the unstandardised average is taken over a cohort whose younger-exposed members carry higher relative risk. The category estimates rise steadily and give no visible sign of a threshold (Figure 1).

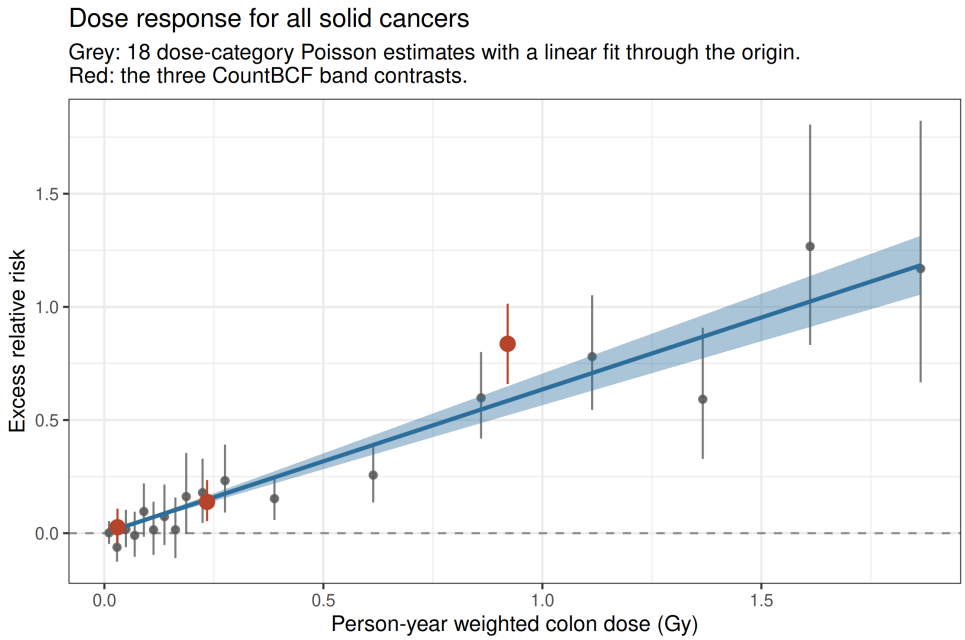



Figure 1: Categorical-dose response from the adjusted parametric model. The estimates increase with dose and do not suggest a threshold over the range considered.

### 1.4.2 Generalised linear model marginal effects

Table 4 reports g-computation marginal effects from adjusted Poisson and negative-binomial models. The Poisson and negative-binomial fits are nearly indistinguishable for all-solid cancer, consistent with the dispersion diagnostics of Section 1.3. Adding dose by modifier interactions improves the all-solid negative-binomial fit by $\Delta\text{AIC} = 54$ (26,931 against 26,985), so heterogeneity is detectable parametrically once the analyst knows to look for it.

Two features of the parametric family are worth carrying forward. The first is the improvement just noted, which establishes that the heterogeneity CountBCF reports in Section 1.6 is present in the data and not a property of the forest. The second is that the unadjusted models are unusable: a minimal negative-binomial model for high-dose stomach returns a rate ratio of 63 with a 95% interval of (0.08, 521), and a minimal negative-binomial model for low-dose all-solid cancer flips to a rate ratio of 0.90 against the Poisson's 0.99. The parametric answer therefore depends entirely on which likelihood and which covariate set the analyst happens to choose, which is precisely the fragility that motivates the present method.

## 1.5 Average effects and agreement with parametric risk models

Each contrast is fitted with two chains, 2,000 burn-in and 1,000 retained draws per chain. The Gelman–Rubin statistic is 1.00 for every contrast, with a maximum of 1.02, and effective sample sizes run from 276 to 1,617 for the exposed-arm effect and from 708 to 1,561 for the ERR. The negative-binomial dispersion posterior is $\theta \approx 157$–$190$ for all-solid cancer (95% credible interval 56 to 585) and $\theta \approx 21$–$30$ for stomach (95% credible interval 8.5 to 77). The all-solid cell counts are thus effectively Poisson once the prognostic forest has done its work, while stomach retains mild genuine overdispersion. This is a sharper answer to the overdispersion question than the generalised-linear-model diagnostics provide, and it is obtained from the same fit that produces the

Table 4: Marginal effects from adjusted parametric models by g-computation. Excess rates are cases per $10^4$ person-years. Intervals are 95% confidence intervals.

| Endpoint | Contrast | Model | Excess rate (95% CI) | Rate ratio (95% CI) |
|---|---|---|---|---|
| All solid | High | Poisson (adjusted) | 32.6 (26.8, 38.3) | 1.532 (1.435, 1.632) |
| | High | NB (adjusted) | 32.7 (27.1, 38.5) | 1.535 (1.440, 1.635) |
| | Medium | NB (adjusted) | 7.35 (3.96, 10.70) | 1.120 (1.064, 1.178) |
| | Low | NB (adjusted) | −0.11 (−2.60, 2.40) | 0.998 (0.958, 1.040) |
| Stomach | High | NB (adjusted) | 5.30 (2.57, 8.14) | 1.319 (1.152, 1.495) |
| | Medium | NB (adjusted) | 1.03 (−0.67, 2.80) | 1.063 (0.961, 1.173) |
| | Low | NB (adjusted) | 0.30 (−1.05, 1.62) | 1.019 (0.940, 1.100) |

causal estimates.

CountBCF and the adjusted negative-binomial generalised linear model agree in every marginal contrast (Table 5 and Figure 2). For high-dose all-solid cancer, CountBCF estimates an excess of 30.2 cases per $10^4$ person-years and a rate ratio of 1.488, compared with 32.7 and 1.535 from the parametric model. The medium-dose contrast is positive under both methods, whereas the low-dose band is uninformative as an overall contrast. CountBCF lies slightly nearer the null throughout, consistent with regularisation of the treatment-effect surface. The exposed-arm version of the estimand is within 0.5% of the population-average version throughout, at 30.37 against 30.22 per $10^4$ person-years for high-dose all-solid cancer, and it is the exposed-arm version that Sections 1.7 and 1.14 use, since a per-Gy risk and an attributable fraction are properties of the exposed.

Table 5: Marginal effects from CountBCF and the adjusted negative-binomial GLM. Excess rates are cases per $10^4$ person-years and ERR equals the rate ratio minus one. Intervals are 95% credible intervals for CountBCF and 95% confidence intervals for the GLM.

| Endpoint | Contrast | Method | Excess rate (95% interval) | ERR (95% interval) | Rate ratio (95% interval) |
|---|---|---|---|---|---|
| All solid | High | CountBCF | 30.2 (23.9, 35.9) | 0.488 (0.381, 0.584) | 1.488 (1.381, 1.584) |
| | High | NB GLM | 32.7 (27.1, 38.5) | 0.535 (0.440, 0.635) | 1.535 (1.440, 1.635) |
| | Medium | CountBCF | 6.95 (3.22, 10.67) | 0.112 (0.051, 0.173) | 1.112 (1.051, 1.173) |
| | Medium | NB GLM | 7.35 (3.96, 10.70) | 0.120 (0.064, 0.178) | 1.120 (1.064, 1.178) |
| | Low | CountBCF | −0.32 (−2.90, 2.26) | −0.005 (−0.046, 0.037) | 0.995 (0.954, 1.037) |
| | Low | NB GLM | −0.11 (−2.60, 2.40) | −0.002 (−0.042, 0.040) | 0.998 (0.958, 1.040) |
| Stomach | High | CountBCF | 4.28 (1.64, 7.11) | 0.252 (0.095, 0.426) | 1.252 (1.095, 1.426) |
| | Medium | CountBCF | 0.22 (−1.58, 2.09) | 0.014 (−0.087, 0.125) | 1.014 (0.913, 1.125) |
| | Low | CountBCF | −0.59 (−2.03, 0.87) | −0.033 (−0.111, 0.051) | 0.967 (0.889, 1.051) |

The high-dose band has a mean dose contrast of 0.9207 Gy relative to the reference group. The per-Gy figure is computed from the exposed-arm standardised ERR, 0.487 (0.386, 0.580), because a risk per Gy is a property of the exposed rather than of the whole cohort; dividing by 0.9207 gives 0.529 per Gy (95% credible interval 0.419, 0.630), reported as 0.53 (0.42, 0.63) per Gy. The population-average standardised ERR of 0.488 (0.381, 0.584) in Table 5 is within half a percent of it on the point estimate and gives 0.53 (0.41, 0.63) on the same rescaling, so nothing turns on the choice; we quote the exposed-arm version throughout for consistency with Sections 1.7 and 1.14. This is a descriptive rescaling of the categorical high-versus-reference contrast, not a separately estimated continuous-dose slope. On this common per-Gy scale it sits among the published values. Preston et al. [2007] report a standardised ERR of 0.47 for the same 1958–1998 follow-up, but with a 90% confidence interval of (0.40, 0.54), so that interval is not directly comparable with ours. The

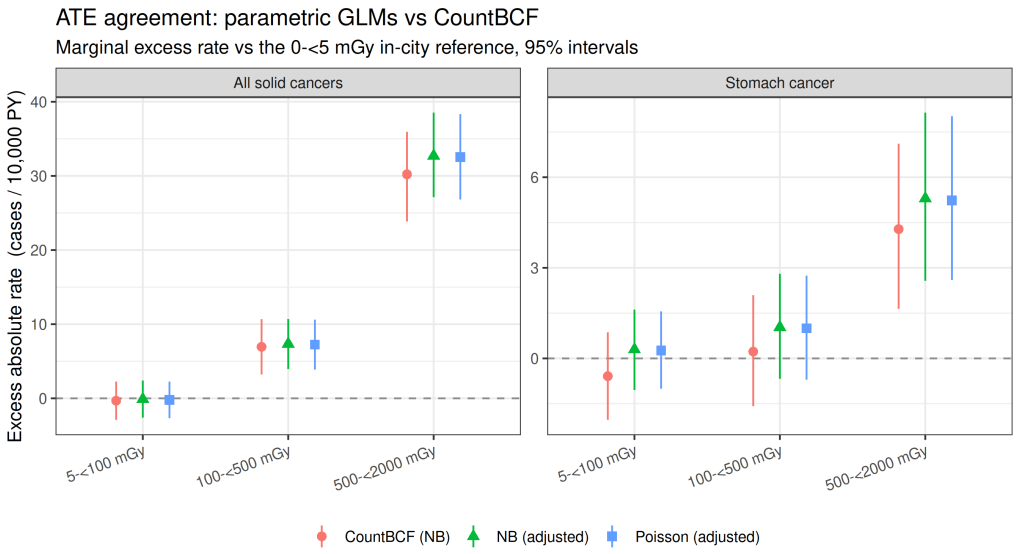



Figure 2: Convergence of marginal effects from CountBCF and the adjusted parametric model. Each estimate is a person-year-standardised excess rate with a 95% interval.

interval-matched comparisons come from the extended 1958–2009 follow-up: Grant et al. [2017] give 0.50 per Gy with a 95% confidence interval of (0.42, 0.59), and French et al. [2017], under internal standardisation, give 0.510 (0.414, 0.612). Our 0.53 (0.42, 0.63) is closer to both of those than the raw comparison with the 90% interval suggests. The medium band gives 0.39 (0.15, 0.65) per Gy on the same rescaling, and for stomach the high band gives 0.26 (0.10, 0.44) per Gy against the published 0.34. The dose response implied by the three band contrasts is consistent with linearity over 0 to 2 Gy: the standardised ERRs of −0.019, 0.092, and 0.487 sit at mean dose contrasts of 0.030, 0.234, and 0.921 Gy, implying per-Gy slopes of roughly −0.63, 0.39, and 0.53, with the low band uninformative because its dose contrast is only 30 mGy (Figure 1). A model given no linearity constraint thus recovers the principal average pattern in the canonical analysis.

Table 6 gives the corresponding E-value sensitivity analysis [VanderWeele and Ding, 2017]. For the high-dose all-solid-cancer contrast, an unmeasured confounder would need associations of at least 2.34 with both dose-band membership and cancer incidence, on the rate-ratio scale, to explain the point estimate away; associations of at least 2.12 with both would be needed to move the interval limit nearest the null to the null. These values are not a proof that unmeasured confounding is absent, but they indicate that a confounder would need to be materially stronger than one weakly related to location-assigned dose. Given that the arms are balanced on every measured demographic except city, and that exposure was determined by where a person happened to be standing on the morning of the bombing, confounders of that strength are hard to name. The medium-dose all-solid result is less robust (E-values 1.41 and 1.23), and the high-dose stomach result is intermediate (1.81 and 1.42). For the low-dose contrasts, the interval E-value is one because the corresponding interval already contains the null. The E-values therefore reinforce the evidence for the high-dose all-solid association while leaving low-dose and site-specific inferences appropriately more tentative.

## 1.6 Heterogeneity of the radiation-effect surface

The main advantage of CountBCF in this application is not a different average effect, but its ability to estimate effect modification without pre-specifying the interaction functions. Throughout this

Table 6: E-value sensitivity analysis for CountBCF. The rate ratios are exposed-arm, person-year-standardised estimates (ATT), which differ slightly from the population-average rate ratios in Table 5. The point and interval E-values give the minimum strength of association, on the rate-ratio scale, that an unmeasured confounder would need with both treatment and outcome to explain the point estimate or move the interval nearest the null to the null.

| Endpoint | Contrast | Exposed-arm RR (95% CrI) | E-value, point | E-value, interval |
|---|---|---|---|---|
| All solid | Low | 0.981 (0.940, 1.023) | 1.16 | 1.00 |
| | Medium | 1.092 (1.035, 1.152) | 1.41 | 1.23 |
| | High | 1.487 (1.386, 1.580) | 2.34 | 2.12 |
| Stomach | Low | 0.956 (0.877, 1.040) | 1.27 | 1.00 |
| | Medium | 1.002 (0.900, 1.112) | 1.04 | 1.00 |
| | High | 1.252 (1.095, 1.426) | 1.81 | 1.42 |

section we call a subgroup difference credible only when the posterior interval for its standardised ERR minus the cohort-wide standardised ATE excludes zero. The footing matters and is fixed here once for the whole document. The standardised ERR is the ratio of person-year-weighted counterfactual rates defined in Section 1.2; it is the quantity a parametric g-computation returns and the one comparable with a Poisson rate ratio and with published case-weighted summaries, and it is therefore the footing on which every subgroup claim in this study is made. The person-year-weighted mean of the per-cell ERR, an average of ratios, runs systematically larger and is reported beside it so that the two can be cross-checked, but it never determines a claim. The EAR is reported to describe the absolute burden and does not determine the separation indicator either. This guards against treating a difference in posterior means as evidence of genuine heterogeneity, and against the two footings being read as though they answered the same question.

A second guard is arithmetic. The separation test is run 294 times across the six endpoint-by-band fits, on nine subgroup partitions per fit. On the standardised footing 34 of those 294 tests flag, and on the per-cell footing 38 do (Table 7). Under independence one would expect about 15 flags at the nominal 5% level, so the flag count is roughly twice chance overall; but the flags are heavily concentrated, with 27 of the 34 falling in the single high-dose all-solid fit and none at all in the three stomach fits. No subgroup claim below rests on the flag alone. A claim is featured only if it survives both the regularisation sweep of Section 1.8 and the data perturbations of Section 1.9; one that survives only one of the two is reported with that qualification attached, as the high-dose sex effect is, and one supported instead by an independent parametric test is identified as such.

Table 7: Multiplicity accounting for the subgroup separation test. Every test run anywhere in the study is counted. A test flags when the 95% credible interval for the subgroup effect minus the cohort average effect excludes zero.

| Fit | Separation tests | Flags, standardised | Flags, per-cell |
|---|---|---|---|
| All solid, 500–< 2000 mGy | 49 | 27 | 26 |
| All solid, 100–< 500 mGy | 49 | 2 | 5 |
| All solid, 5–< 100 mGy | 49 | 5 | 7 |
| Stomach, 500–< 2000 mGy | 49 | 0 | 0 |
| Stomach, 100–< 500 mGy | 49 | 0 | 0 |
| Stomach, 5–< 100 mGy | 49 | 0 | 0 |
| Total | 294 | 34 | 38 |

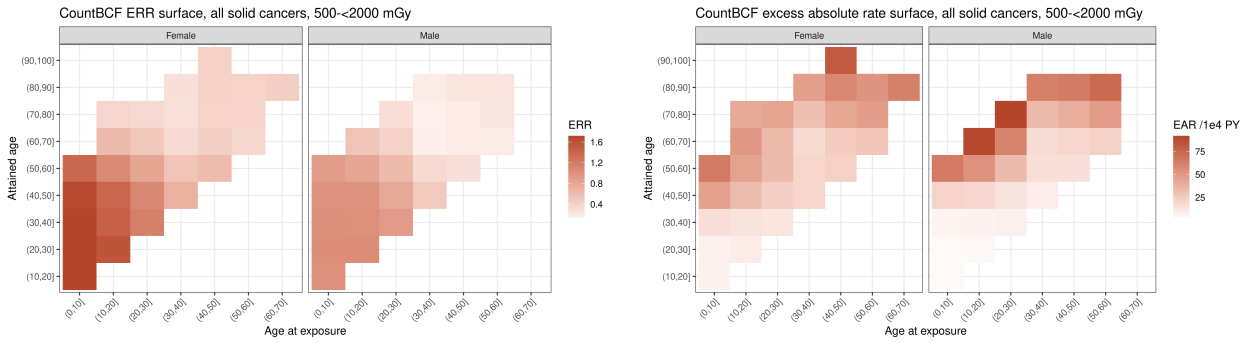
### 1.6.1 How much heterogeneity is present at all

Before examining named subgroups it is useful to ask how much variation the posterior surface contains. Table 8 reports the posterior person-year-weighted standard deviation of the per-cell ERR across exposed cells, together with the share of exposed cells whose effect is positive and the share whose 95% interval excludes zero.

Table 8: Cell-level heterogeneity of the CountBCF effect surface. The first column is the posterior person-year-weighted standard deviation of the per-cell ERR across exposed cells.

| Endpoint | Contrast | SD of per-cell ERR (95% CrI) | Cells with ERR $> 0$ | Cells with CrI excluding 0 |
|---|---|---|---|---|
| All solid | High | 0.584 (0.416, 0.773) | 98.6% | 63.0% |
| | Medium | 0.253 (0.198, 0.324) | 82.8% | 2.0% |
| | Low | 0.233 (0.183, 0.299) | 58.1% | 2.4% |
| Stomach | High | 0.004 (0.000, 0.030) | 100% | 100% |
| | Medium | 0.146 (0.076, 0.240) | 73.5% | 0% |
| | Low | 0.186 (0.135, 0.257) | 54.3% | 0% |

For all-solid cancer at high dose the heterogeneity is large and credible: the per-cell ERR has a posterior median of 0.60 with a tenth-to-ninetieth percentile range of 0.16 to 1.57, and nearly two thirds of individual cells have an interval excluding zero. At medium and low dose the surface is much flatter and almost no individual cell separates, which is a statement about power rather than about the absence of modification. For stomach at high dose the moderating forest returns an essentially constant effect, an ERR of 0.252 in every subgroup with a posterior standard deviation of 0.004; that case is discussed in Section 1.6.4. Figures 3a and 3b display the fitted relative and absolute surfaces over the covariate space.




(a) Relative effect (ERR).

(b) Absolute effect (EAR).

Figure 3: Estimated CountBCF effect surfaces for high-dose all-solid cancer over the observed covariate space. The relative and absolute surfaces move in opposite directions with attained age.

### 1.6.2 All solid cancers at 500–< 2000 mGy

Table 9 gives the complete set of high-dose subgroup effects with their separation tests. Five conclusions follow.

First, age at exposure is the dominant modifier of the relative effect. The standardised ERR declines more than fourfold, from 1.26 (0.85, 1.71) among those exposed at ages 0–9 to 0.29 (0.14,

0.44) among those exposed at age 40 or later, and both tails separate credibly from the average. This is the textbook LSS result, recovered with no interaction term specified anywhere.

Second, attained age modifies the relative and absolute effects in opposite directions. The standardised ERR falls from 1.40 (0.97, 1.84) below age 40 to 0.30 (0.11, 0.51) at age 80 or older, while the EAR rises from 8.7 to 57.3 excess cases per $10^4$ person-years, and every attained-age band separates. The two are projections of a single surface, since the EAR is approximately the baseline rate multiplied by the ERR and the baseline rate climbs faster with attained age than the relative effect fades.

Third, time since exposure moves the absolute effect but not the relative one. The ERR is statistically flat across 20 to more than 40 years of follow-up, with no subgroup separating, while the EAR nearly quadruples from 15.2 to 57.3 per $10^4$ person-years. The relative excess established early does not wash out; the absolute burden grows because the baseline does.

Fourth, sex separates at high dose, with a standardised ERR of 0.58 (0.45, 0.72) in women against 0.38 (0.25, 0.51) in men, a female-to-male ERR ratio of 1.54. That is close to the published 1.6, and the agreement is closer still on the exposed-cell footing used in Section 1.7, where the ratio is $0.603/0.369 = 1.63$. This dimension is the clearest illustration in the study of why the footing has to be fixed in advance: the same contrast computed as an average of per-cell ratios gives 0.94 against 0.74, a ratio of 1.27, and does not separate, which would have understated our own agreement with the canonical model. Two qualifications keep the sex effect out of the featured results even so. Section 1.8 shows that its magnitude is strongly sensitive to treatment-effect regularisation, collapsing ninefold under a halved leaf concentration even though its interval never covers zero, and Section 1.9 shows that it is carried by breast and thyroid, its interval covering zero once those two sites are removed. We also note that it reverses a finding from a preliminary analysis on a 30% subsample, which had the male ERR marginally above the female one.

Fifth, the two cities are indistinguishable at high dose, at 0.49 (0.37, 0.60) in Hiroshima against 0.48 (0.31, 0.67) in Nagasaki, which is what one would want if the DS02 weighted organ doses have reconciled the Hiroshima neutron component correctly. It is also the fact that constrains the reading of Section 1.10: any city difference confined to the lowest band has to be reconciled with exact agreement where the dose signal is strongest.

These patterns match the central qualitative findings of LSS incidence modelling: relative risk is greater following younger-age exposure, while the absolute rate burden grows at older attained ages [Preston et al., 2007, Grant et al., 2017].

### 1.6.3 All solid cancers at 100–< 500 mGy

On the standardised footing no subgroup separates at 100–< 500 mGy. The sex contrast comes closest, with a male standardised ERR of 0.058 (−0.021, 0.143) and a difference from the average of −0.054 (−0.108, 0.003), against a female 0.156 (0.079, 0.240) and +0.045 (−0.002, 0.093); both intervals just cover zero. On the per-cell footing the same contrast is 0.061 against 0.222 and does flag, at −0.101 (−0.200, −0.002) and +0.059 (0.001, 0.118), which is a difference of footing rather than of evidence: the medium-dose sex effect is borderline either way and is not featured. No age-at-exposure or attained-age subgroup separates at this dose on either footing.

The pattern across dose bands is nevertheless informative, and it points at something the canonical model already contains. The female-to-male ratio of standardised ERRs is about 2.7 at 100–< 500 mGy and 1.54 at 500–< 2000 mGy, so the sex gap is widest where the male response is weakest in absolute terms. That is what the published curvature of the male dose response implies. Grant et al. [2017] report upward curvature in the male dose response, including an apparent flattening of the response between roughly 0.2 and 0.7 Gy; French et al. [2017] report strong

Table 9: High-dose all-solid-cancer conditional effects (500–< 2000 mGy against the 0–< 5 mGy reference). The first three numeric columns are on the standardised footing, which is the one that determines every claim: a subgroup is marked as separated when the 95% posterior interval for its standardised ERR minus the cohort-wide standardised ATE of 0.488 excludes zero. The final column is the person-year-weighted mean of the per-cell ERR, an average of ratios, given for cross-reference only; it runs systematically larger and is taken against a different reference, the per-cell ATE of 0.87. The Cases column counts cases in the subgroup across both dose arms.

| Dimension | Subgroup | Cases | Std. ERR (95% CrI) | CATE − ATE, std. (95% CrI) | Sep.? | Per-cell ERR |
|---|---|---|---|---|---|---|
| Sex | Male | 3,089 | 0.38 (0.25, 0.51) | −0.11 (−0.21, −0.02) | Yes | 0.74 |
| | Female | 3,662 | 0.58 (0.45, 0.72) | +0.09 (0.01, 0.18) | Yes | 0.94 |
| Age at exposure | 0–9 | 532 | 1.26 (0.85, 1.71) | +0.77 (0.39, 1.18) | Yes | 1.45 |
| | 10–19 | 1,303 | 0.80 (0.60, 1.02) | +0.31 (0.14, 0.50) | Yes | 1.08 |
| | 20–29 | 1,159 | 0.51 (0.33, 0.71) | +0.02 (−0.15, 0.19) | No | 0.72 |
| | 30–39 | 1,499 | 0.27 (0.10, 0.45) | −0.22 (−0.37, −0.06) | Yes | 0.37 |
| | 40+ | 2,258 | 0.29 (0.14, 0.44) | −0.20 (−0.32, −0.08) | Yes | 0.32 |
| Attained age | < 40 | 167 | 1.40 (0.97, 1.84) | +0.91 (0.52, 1.33) | Yes | 1.41 |
| | 40–59 | 1,677 | 0.91 (0.71, 1.12) | +0.42 (0.27, 0.59) | Yes | 1.00 |
| | 60–79 | 3,818 | 0.33 (0.22, 0.44) | −0.16 (−0.23, −0.09) | Yes | 0.36 |
| | 80+ | 1,089 | 0.30 (0.11, 0.50) | −0.19 (−0.35, −0.02) | Yes | 0.32 |
| City | Hiroshima | 4,369 | 0.49 (0.37, 0.60) | +0.004 (−0.07, 0.08) | No | 0.77 |
| | Nagasaki | 2,382 | 0.48 (0.31, 0.67) | −0.007 (−0.14, 0.13) | No | 1.02 |
| Time since exposure | < 20 y | 939 | 0.45 (0.29, 0.62) | −0.04 (−0.16, 0.10) | No | 0.93 |
| | 20–29 y | 1,430 | 0.44 (0.32, 0.56) | −0.05 (−0.13, 0.03) | No | 0.90 |
| | 30–39 y | 1,670 | 0.49 (0.38, 0.60) | +0.002 (−0.04, 0.04) | No | 0.88 |
| | 40+ y | 2,712 | 0.53 (0.39, 0.66) | +0.04 (−0.05, 0.13) | No | 0.75 |

Excess absolute rates per $10^4$ person-years for the same subgroups, in the order listed, are 28.9 and 31.0 by sex; 24.6, 34.3, 31.8, 24.4, and 35.5 by age at exposure; 8.7, 34.3, 37.6, and 57.3 by attained age; 33.0 and 26.2 by city; and 15.2, 20.8, 32.1, and 57.3 by time since exposure.

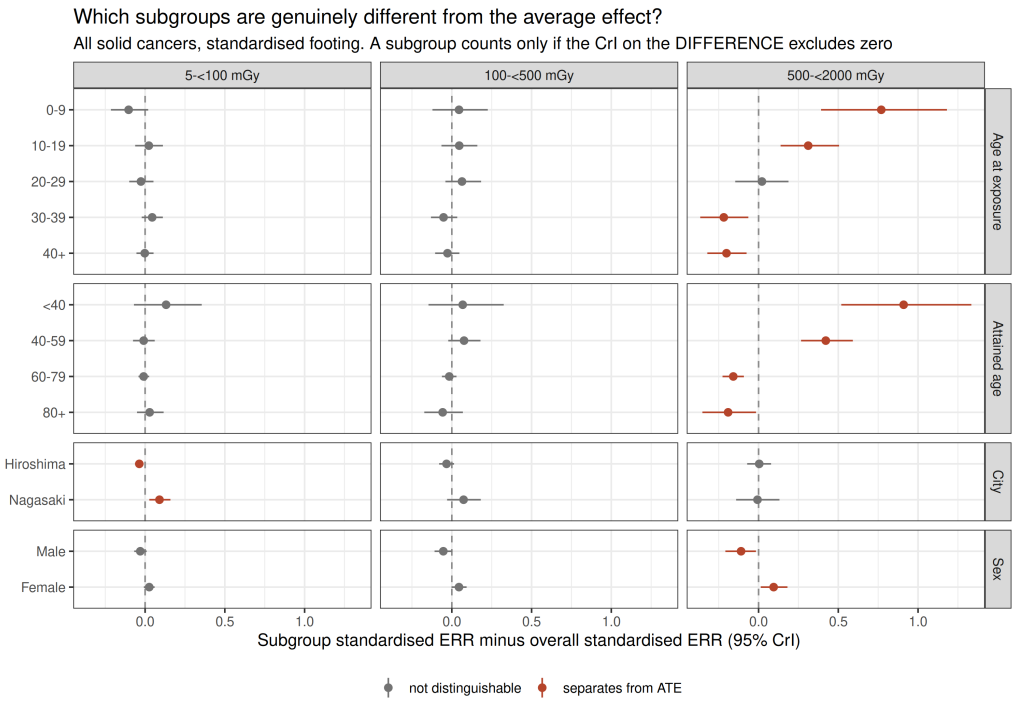



Figure 4: Subgroup separation for the high-dose all-solid contrast. Each point is the posterior mean of the subgroup conditional effect minus the population average effect (ATE), with its 95% credible interval. Intervals excluding zero identify credible effect modification.

evidence of a curvilinear dose response among males, at $P = 0.008$ under internal standardisation; and Preston et al. [2007] already reported a borderline sex-averaged test for quadratic departure from linearity over 0–2 Gy, at $P = 0.09$, with an estimated curvature of 0.3 (90% CI 0.01, 0.90). A male response that is flat across the medium band and then rises steeply depresses the male ERR at 100–< 500 mGy relative to a linear expectation and restores it at 500–< 2000 mGy, which is exactly the band pattern above. CountBCF was given no dose-shape assumption of any kind, so recovering the signature of a curvilinear male response from the band contrasts alone is a further piece of convergent evidence rather than a discrepancy.

### 1.6.4 Stomach cancer as a counterpoint

Stomach cancer provides a useful counterpoint. Its high-dose average effect is positive, but its estimated ERR surface is essentially constant across the examined covariates, with a posterior standard deviation of 0.004. This is not a failure of the estimator; it is the differential-shrinkage prior doing what it is designed to do. With 269 exposed stomach cases the data cannot support effect modification, and the published tests agree, giving $P = 0.4$ for the stomach age-at-exposure trend and $P = 0.05$ for the attained-age effect. Where the data do not support a complex surface, the fit shrinks toward a homogeneous effect rather than creating spurious subgroup differences. The all-solid regularisation sweep tests whether this behavior could be excessive, and the stomach-specific repetition of that grid is reported in Table 12. It gives the same null result under the ordinary settings and identifies the extreme shared-prior fit as unstable rather than as evidence of stomach effect modification.

## 1.7 Comparison with the canonical LSS surface

The average-effect comparison is only a partial validation. We therefore evaluate the CountBCF surface against the refitted risk model of Section 1.4.1 at the same person-year cells. Two comparisons are possible. The standardised comparison weights each surface by the same CountBCF-estimated control-arm rate, which makes it the analogue of a rate ratio and the fair test. The per-cell comparison is the plain person-year-weighted mean of the cell ERRs; it is systematically larger for both surfaces and is what a naive comparison would produce. We report the standardised comparison. Unlike the separation tests of Section 1.6, which standardise the conditional effect over the whole cohort, this comparison is evaluated only at the exposed person-year cells, because the parametric ERR is proportional to dose and collapses to zero at the reference cells whereas the CountBCF contrast does not; the two surfaces can therefore only be compared where the exposure actually occurred. A given subgroup's CountBCF ERR here can accordingly differ by a few percent from its cohort-standardised value in Table 9.

### 1.7.1 All solid cancers

On the standardised scale the two surfaces agree in all but two of 28 high-dose subgroups, all but three of 28 medium-dose subgroups, and all but six of 28 low-dose subgroups (Table 10 and Figure 5). Their person-year-weighted mean EAR is identical to four significant figures at high dose, 30.37 per $10^4$ person-years for each model. Across the 5,681 exposed cells the two surfaces correlate at 0.57 on the Pearson scale and 0.80 on the Spearman scale; the rank correlation is the higher of the two, so the earlier description of 0.57 as a rank correlation understated the agreement. The agreement is notable because CountBCF was supplied neither a linear dose term nor any interaction involving age, sex, or city. A nonparametric causal forest, given no functional form and no interaction terms, reproduces the canonical parametric risk surface for all solid cancers over almost the whole covariate space.

Table 10: Standardised comparison of the CountBCF surface with the refitted canonical risk model, evaluated at the same person-year cells. Every subgroup whose 95% credible interval for the difference excludes zero is listed, for all three dose bands, together with the overall row and the two sexes at high dose for context. Rows marked † are the departures.

| Contrast | Subgroup | Refitted model ERR | CountBCF ERR (95% CrI) | Difference (95% CrI) |
|---|---|---|---|---|
| High | Overall | 0.553 | 0.487 (0.386, 0.580) | −0.066 (−0.168, 0.026) |
| | Attained age < 40 † | 2.759 | 1.330 (0.931, 1.758) | −1.429 (−1.863, −0.965) |
| | Time since exposure < 20 y † | 0.659 | 0.471 (0.301, 0.640) | −0.188 (−0.356, −0.015) |
| | Age at exposure 0–9 | 1.522 | 1.184 (0.794, 1.607) | −0.338 (−0.735, 0.090) |
| | Female | 0.707 | 0.603 (0.465, 0.741) | −0.104 (−0.243, 0.035) |
| | Male | 0.396 | 0.369 (0.240, 0.500) | −0.027 (−0.157, 0.104) |
| Medium | Overall | 0.138 | 0.092 (0.035, 0.152) | −0.046 (−0.103, 0.013) |
| | Age at exposure 0–9 † | 0.368 | 0.114 (−0.053, 0.294) | −0.253 (−0.421, −0.073) |
| | Attained age < 40 † | 0.736 | 0.142 (−0.072, 0.389) | −0.593 (−0.813, −0.340) |
| | Female exposed at 0–9 † | 0.508 | 0.193 (−0.016, 0.440) | −0.315 (−0.531, −0.070) |
| Low | Overall | 0.019 | −0.019 (−0.060, 0.023) | −0.038 (−0.078, 0.005) |
| | Hiroshima † | 0.019 | −0.043 (−0.089, 0.003) | −0.063 (−0.108, −0.016) |
| | Nagasaki | 0.017 | 0.088 (0.006, 0.171) | +0.072 (−0.011, 0.154) |
| | Age at exposure 0–9 † | 0.044 | −0.141 (−0.256, −0.015) | −0.185 (−0.299, −0.060) |
| | Time since exposure 40+ y † | 0.018 | −0.059 (−0.115, −0.001) | −0.077 (−0.134, −0.019) |
| | Male exposed at 0–9 † | 0.031 | −0.143 (−0.279, 0.010) | −0.174 (−0.309, −0.022) |
| | Female exposed at 0–9 † | 0.059 | −0.136 (−0.274, 0.015) | −0.195 (−0.333, −0.045) |
| | Male exposed at 20–29 † | 0.013 | −0.110 (−0.222, 0.011) | −0.123 (−0.235, −0.002) |

The low-dose block deserves a separate remark, because it is the one that bears on Section 1.10. Six of the 28 low-dose subgroups depart from the canonical surface on the standardised scale, and every one of them departs downward: CountBCF puts the low-band ERR below what a linear dose response extrapolated from the whole 0–2 Gy range predicts. Hiroshima is among the six, at −0.063 (−0.108, −0.016). Nagasaki is not, at +0.072 (−0.011, 0.154). On the scale that this study uses for every other claim, therefore, Nagasaki's low-band ERR is not distinguishable from what the canonical model already predicts, and Hiroshima's is. Whatever the low-dose city contrast of Section 1.10 is, it is at least as much a statement about Hiroshima sitting below the canonical prediction as about Nagasaki sitting above it.

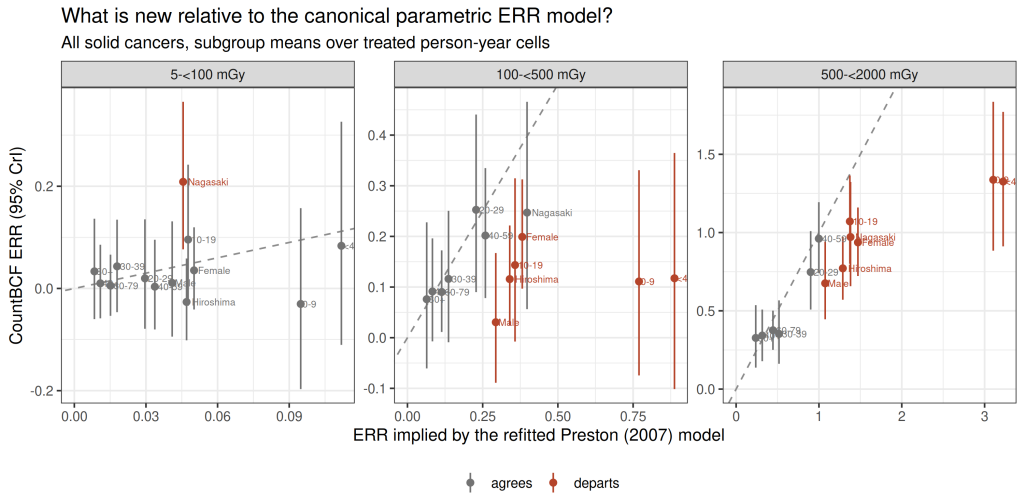



Figure 5: CountBCF conditional ERRs compared with the refitted canonical risk model. The models agree across most of the observed covariate space; the largest differences are confined to young attained age and short latency.

The disagreements are localized rather than scattered. They occur at young attained age and short time since exposure, where the parametric attained-age power law extrapolates most strongly. At attained age below 40 the refitted model predicts a high-dose ERR of 2.76 at 0.92 Gy, while CountBCF estimates 1.33 (0.93, 1.76) and excludes the parametric value. The same pattern appears one dose band down, at 0.74 against 0.14, and for stomach, at 2.89 against 0.25. A power law in attained age is forced to be monotone and unbounded as attained age tends to zero; the data in that corner are sparse, with only 51 exposed cases below attained age 40 in the high band, and CountBCF's answer is a flattening rather than a divergence. The published analysis itself flags this region as unsettled, noting that a more complex pattern provides a somewhat better fit for all solid cancers and that future analyses should give more attention to alternative descriptions of age-at-exposure effects on both the ERR and the EAR [Preston et al., 2007]. The competing explanation has to be stated plainly: because the moderating-forest prior shrinks toward a constant, a flattening in a thin corner is also what excessive shrinkage would produce. Section 1.8 is the decisive test of that alternative.

### 1.7.2 Stomach cancer

For stomach at high dose, 21 of the 28 subgroups depart on the standardised scale, and 23 of 28 on the per-cell scale, because the parametric model carries effect modification, with an ERR proportional to attained age to the power −1.5, a decline of 13% per decade of age at exposure,

and a female-to-male ratio of 2.3, while CountBCF returns a flat ERR of 0.252. The two are not equally supported by the data. The published significance tests for stomach give $P = 0.4$ for the age-at-exposure trend and $P = 0.05$ for attained age, so the modification being fitted is at or beyond the edge of detectability and its estimates are borrowed from the functional form rather than read off the stomach data. CountBCF, which must find modification in the stomach data alone, finds none. The right reading is not that either analysis is wrong about stomach, but that the apparent stomach effect modification in the canonical analysis is a property of the model class rather than of the stomach data. What the discrepancy identifies is the assumption doing the work: CountBCF requires the site-specific data to support a modifier, whereas the parametric estimate extrapolates a specified functional form. The corresponding regularisation question is now checked directly rather than left to the single default fit. The twelve-setting stomach sweep reported in Section 1.8, Table 12, finds no subgroup separation on either ERR footing at the default and all but one of the settings; the exceptional setting produces an unstable, highly heterogeneous surface rather than a credible stomach-specific pattern.

## 1.8 Robustness to regularisation of the moderating forest

Because CountBCF deliberately tightens the prior on the treatment-effect surface in order to suppress spurious heterogeneity, any featured subgroup effect must be shown to survive both loosening and tightening that prior. Table 11 reports twelve refits of the featured contrast, all solid cancers at 500–< 2000 mGy, varying the number of moderating trees, the tree-depth penalty on the moderating forest, and the overall leaf concentration. Figure 6 displays the same sweep.

Table 11: Sensitivity of the featured contrast (all solid cancers, 500–< 2000 mGy) to regularisation of the moderating forest. All entries are on the standardised footing; the last two columns are posterior differences with 95% credible intervals. $m_\tau$ and the depth prior act on the moderating forest alone, whereas the leaf concentration $a_0$ is shared by both forests, so the $a_0$ rows are not a moderating-forest-only sweep. The twelve rows are ten distinct fits: the default specification is $m_\tau = 50$ with depth prior $(0.25, 3)$ and $a_0 \times 1$, so it is the common row of all three sweeps and is reported three times (‡).

| Specification | Standardised ERR (95% CrI) | Std. ERR, exposed < 20 minus 40+ | Std. ERR, female minus male |
|---|---|---|---|
| $m_\tau = 10$ (tightest) | 0.434 (0.345, 0.521) | 0.421 (0.208, 0.612) | 0.156 (0.003, 0.307) |
| $m_\tau = 25$ | 0.486 (0.389, 0.582) | 0.529 (0.298, 0.787) | 0.225 (0.051, 0.407) |
| $m_\tau = 50$ (default) ‡ | 0.495 (0.401, 0.594) | 0.527 (0.281, 0.778) | 0.256 (0.074, 0.431) |
| $m_\tau = 100$ | 0.508 (0.416, 0.608) | 0.597 (0.344, 0.864) | 0.223 (0.038, 0.414) |
| $m_\tau = 200$ (loosest) | 0.505 (0.405, 0.605) | 0.604 (0.356, 0.859) | 0.269 (0.084, 0.456) |
| Depth prior (0.05, 4), tight | 0.494 (0.400, 0.599) | 0.563 (0.313, 0.813) | 0.208 (0.029, 0.380) |
| Depth prior (0.25, 3), default ‡ | 0.495 (0.401, 0.594) | 0.527 (0.281, 0.778) | 0.256 (0.074, 0.431) |
| Depth prior (0.5, 2), loose | 0.496 (0.398, 0.598) | 0.574 (0.325, 0.824) | 0.240 (0.054, 0.438) |
| Depth prior (0.95, 2), prognostic-like | 0.489 (0.397, 0.585) | 0.583 (0.338, 0.839) | 0.260 (0.086, 0.445) |
| $a_0 \times 0.5$ (both forests) | 0.465 (0.359, 0.576) | 0.272 (0.125, 0.422) | 0.029 (0.003, 0.057) |
| $a_0 \times 1$ (default) ‡ | 0.495 (0.401, 0.594) | 0.527 (0.281, 0.778) | 0.256 (0.074, 0.431) |
| $a_0 \times 2$ (both forests) | 0.463 (0.367, 0.558) | 0.579 (0.322, 0.831) | 0.254 (0.061, 0.436) |

Three conclusions follow. First, the average effect is essentially prior-free: the standardised ERR sits between 0.434 and 0.508 across a twentyfold change in the prior variance of the moderating surface and across every depth prior tried, so the headline number is not an artefact of the shrinkage. Second, the age-at-exposure gradient survives everything. The standardised ERR difference between those exposed before age 20 and after age 40 has a credible interval excluding zero in all twelve specifications, including the tightest moderating forest in the sweep, which gives 0.421 (0.208, 0.612), and the halved leaf concentration, which gives 0.272 (0.125, 0.422). Its magnitude

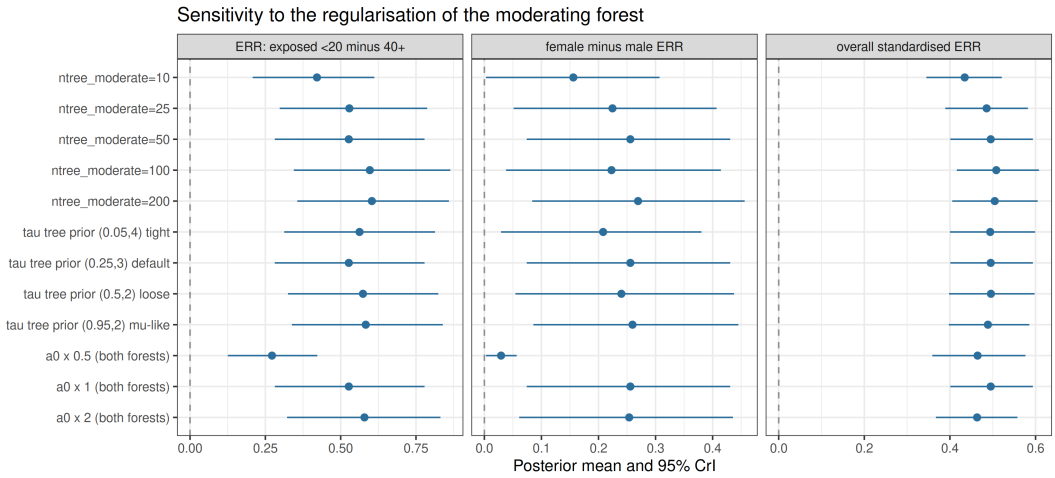



Figure 6: Sensitivity of the featured contrast to the moderating-forest prior. All three panels are on the standardised footing, matching Table 11.

scales with prior looseness, from 0.27 to 0.60, a range of roughly 2.2-fold, so the size of the gradient is prior-dependent while its existence is not.

Third, the sex difference behaves differently, and the distinction is worth stating carefully because it is exactly the sort of claim the choice of footing decides. On the standardised footing the female-minus-male difference also has a credible interval excluding zero in all twelve specifications, so its existence is no less prior-robust than the gradient's. What is not robust is its size: it falls from 0.256 (0.074, 0.431) at the defaults to 0.029 (0.003, 0.057) when the shared leaf concentration is halved, a ninefold collapse against the gradient's 2.2-fold range. A quantity whose interval never covers zero but whose magnitude moves by an order of magnitude under a prior change is evidence that something is present, not evidence about how large it is, and the second is what a featured subgroup claim would have to assert. The sex effect therefore stays out of the featured results, but for the reason established in Section 1.9 rather than this one: it is carried by breast and thyroid, and its interval covers zero once those two sites are removed.

Finally, the Nagasaki-minus-Hiroshima standardised ERR difference at high dose covers zero in eleven of the twelve specifications and never exceeds 0.06 in magnitude in any of them, confirming the null reported in Section 1.6.2. The exception is the halved leaf concentration, at +0.029 (0.004, 0.054), and it illustrates the point just made about that setting rather than contradicting the null: halving $a_0$ shrinks the whole moderating surface so hard that every subgroup contrast collapses to a value near 0.03 with an interval only about 0.05 wide, which is why the sex difference separates there at 0.029 (0.003, 0.057) as well. Separation under that prior reflects how tightly the posterior is determined, not how large the underlying contrast is.

The same twelve-setting grid was then applied to the stomach high-dose contrast to test the mirror-image concern: could a modifier hidden by the default prior appear when the regularisation is changed? Table 12 reports the result. At the default, the posterior person-year-weighted standard deviation of the stomach ERR surface is 0.0042, none of the 49 subgroup tests separates on either the standardised or per-cell footing, and the largest standardised subgroup deviation from the average is 0.0002. The same zero separation result holds in the other eleven settings except for the extreme $a_0 \times 2$ setting, where the surface becomes unstable: its standardised heterogeneity is 1.14 and 9 of 49 standardised tests (15 of 49 per-cell tests) separate. The overall effect also shifts to 0.184 (0.036, 0.356) there. The ordinary regularisation changes therefore do not reveal hidden stomach

heterogeneity; the extreme shared prior produces a pathological surface and is not evidence for a stomach modifier.

Table 12: Sensitivity of the stomach high-dose contrast to the regularisation grid used for the all-solid analysis. The standardised ERR is the person-year-weighted exposed-arm contrast with its 95% credible interval. The next two columns give the posterior person-year-weighted standard deviation of the per-cell ERR surface and the largest absolute subgroup deviation from the average, both on the standardised footing. The final columns count subgroup tests whose 95% interval excludes zero, out of 49 tests, on the standardised and per-cell footings respectively. The $a_0$ rows act on both forests; the default is repeated across the three sweeps (‡).

| Specification | Standardised ERR (95% CrI) | SD(ERR) | Max $\lvert d \rvert$ | Std. flags / 49 | Per-cell flags / 49 |
|---|---|---|---|---|---|
| $m_\tau = 10$ (tightest) | 0.226 (0.078, 0.390) | 0.0007 | 0.00004 | 0 | 0 |
| $m_\tau = 25$ | 0.242 (0.084, 0.417) | 0.0026 | 0.00014 | 0 | 0 |
| $m_\tau = 50$ (default) ‡ | 0.253 (0.078, 0.429) | 0.0042 | 0.00019 | 0 | 0 |
| $m_\tau = 100$ | 0.255 (0.082, 0.427) | 0.0075 | 0.00038 | 0 | 0 |
| $m_\tau = 200$ (loosest) | 0.260 (0.089, 0.452) | 0.0124 | 0.00063 | 0 | 0 |
| Depth prior $(0.05, 4)$, tight | 0.252 (0.080, 0.422) | 0.0008 | 0.00010 | 0 | 0 |
| Depth prior $(0.25, 3)$, default ‡ | 0.253 (0.078, 0.429) | 0.0042 | 0.00019 | 0 | 0 |
| Depth prior $(0.5, 2)$, loose | 0.253 (0.088, 0.416) | 0.0090 | 0.00061 | 0 | 0 |
| Depth prior $(0.95, 2)$, extreme | 0.248 (0.087, 0.419) | 0.0414 | 0.00655 | 0 | 0 |
| $a_0 \times 0.5$ (both forests) | 0.234 (0.085, 0.397) | 0.0754 | 0.053 | 0 | 0 |
| $a_0 \times 1$ (default) ‡ | 0.253 (0.078, 0.429) | 0.0042 | 0.00019 | 0 | 0 |
| $a_0 \times 2$ (both forests) | 0.184 (0.036, 0.356) | 1.136 | 1.668 | 9 | 15 |

## 1.9 Robustness to the data construction

Table 13 reports fourteen refits of the featured contrast, each changing one element of the data construction, with Figure 7 showing the same fits.

Table 13: Sensitivity of the featured contrast to the data construction. Each row changes one element relative to the baseline. All ERR entries are on the standardised footing fixed in Section 1.6; the last two columns are posterior differences with 95% credible intervals.

| Perturbation | Cells | Cases | Standardised ERR (95% CrI) | EAR/$10^4$ PY | Std. ERR, < 20 minus 40+ | Std. ERR, female minus male |
|---|---|---|---|---|---|---|
| Baseline (all solid, colon bands, NB) | 8,249 | 6,751 | 0.495 (0.401, 0.594) | 30.7 | 0.527 (0.281, 0.778) | 0.256 (0.074, 0.431) |
| Poisson likelihood | 8,249 | 6,751 | 0.498 (0.402, 0.598) | 30.8 | 0.563 (0.311, 0.819) | 0.240 (0.050, 0.413) |
| Not in city added to control arm | 9,534 | 10,745 | 0.501 (0.408, 0.603) | 30.9 | 0.552 (0.299, 0.799) | 0.233 (0.059, 0.417) |
| Reference widened to 0–< 100 mGy | 14,222 | 11,157 | 0.503 (0.406, 0.606) | 31.0 | 0.567 (0.334, 0.807) | 0.237 (0.075, 0.411) |
| High band extended to include $\geq$ 2 Gy | 10,714 | 6,936 | 0.588 (0.494, 0.691) | 35.9 | 0.713 (0.456, 0.986) | 0.289 (0.096, 0.480) |
| Excluding breast and thyroid | 8,249 | 6,137 | 0.405 (0.310, 0.510) | 23.1 | 0.287 (0.049, 0.534) | 0.066 (−0.112, 0.261) |
| Excluding non-melanoma skin | 8,249 | 6,605 | 0.479 (0.384, 0.576) | 29.0 | 0.491 (0.253, 0.752) | 0.210 (0.035, 0.398) |
| Excluding stomach | 8,249 | 4,975 | 0.575 (0.458, 0.697) | 25.9 | 0.532 (0.249, 0.825) | 0.206 (−0.030, 0.412) |
| Non-sex-specific solid only | 8,249 | 5,502 | 0.525 (0.416, 0.642) | 26.5 | 0.420 (0.161, 0.689) | 0.142 (−0.069, 0.336) |
| Stomach, colon-dose bands | 8,249 | 1,776 | 0.253 (0.078, 0.429) | 4.41 | −0.0001 (−0.003, 0.002) | 0.00002 (−0.0004, 0.0005) |
| Stomach, stomach-dose bands | 8,075 | 1,774 | 0.256 (0.092, 0.443) | 4.45 | −0.0001 (−0.003, 0.002) | 0.00001 (−0.0004, 0.0005) |
| Lung, lung-dose bands | 7,636 | 683 | 0.666 (0.356, 1.001) | 4.21 | −0.047 (−0.472, 0.272) | 0.480 (0.002, 1.086) |
| Colon, colon-dose bands | 8,249 | 584 | 0.297 (0.019, 0.586) | 1.64 | 0.052 (−0.211, 0.335) | −0.095 (−0.473, 0.242) |
| Liver, liver-dose bands | 7,917 | 592 | 0.291 (0.024, 0.609) | 1.66 | 0.073 (−0.185, 0.396) | −0.121 (−0.455, 0.193) |

Not-in-city survivors change nothing. Adding the 1,285 not-in-city cells to the control arm with an indicator in the prognostic forest moves the standardised ERR from 0.495 to 0.501, a change of 1.2% that is well inside the Monte Carlo noise. This reproduces the published conclusion that the dose-response parameter estimates are virtually unchanged when the not-in-city group is excluded, and it answers the design question raised in Section 1.2: the indicator works, and the choice does not matter much either way.

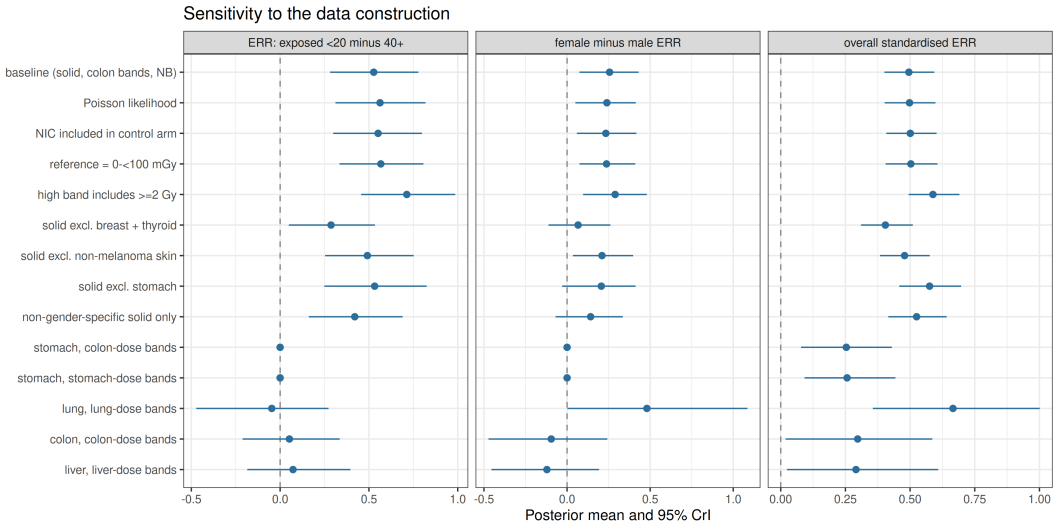



Figure 7: Sensitivity of the featured contrast to the data construction. All three panels are on the standardised footing, matching Table 13. The average effect and the age-at-exposure gradient hold their sign and separation under every perturbation, the largest movements being the removal of breast and thyroid, which lowers both, and the extension of the high band beyond 2 Gy, which raises both. The sex difference behaves differently, losing its separation from zero in three of the four all-solid composition variants.

The reference-band and dose-cap definitions are second-order. Widening the reference group to 0–< 100 mGy gives 0.503 against 0.495, and restoring the survivors at or above 2 Gy raises the estimate to 0.588, which is exactly what a dose response predicts, since it raises the exposed arm's mean dose; on a per-Gy scale the two agree. The likelihood does not matter either, with Poisson and negative binomial giving 0.498 and 0.495, consistent with the near-Poisson dispersion posterior of Section 1.5.

Cancer-site composition is the one perturbation that matters, and it settles the sex question left open in Section 1.8. Removing breast and thyroid drops the standardised ERR from 0.495 to 0.405, a fall of 18%, and the standardised female-minus-male difference collapses from 0.256 (0.074, 0.431) to 0.066 (−0.112, 0.261), its interval now covering zero. Removing non-melanoma skin, by contrast, changes almost nothing, giving 0.479 with an age-at-exposure gradient of 0.491 against the baseline 0.527, so non-melanoma skin is not the driver of the aggregate's age-at-exposure shape despite being the site with the steepest age-at-exposure effect in the published heterogeneity tests. The sex effect in the all-solid aggregate is thus carried by the female-only sites, which matches the published finding that the sex difference disappears when the analysis is restricted to non-sex-specific cancers on the EAR scale. Our non-sex-specific fit leaves a difference of 0.142 (−0.069, 0.336), attenuated by about 45% and no longer separating from zero, although the point estimate stays positive in the direction implied by the published ERR sex ratio of 1.8 for non-sex-specific cancers.

This is what disqualifies the sex difference from the featured results. Its credible interval excludes zero under every prior in Table 11, but it does not survive the data: three of the four all-solid composition variants extinguish it, the exception being the removal of non-melanoma skin. The contrast with the age-at-exposure gradient is the point. That gradient survives every data perturbation for all-solid cancer, with a credible interval excluding zero in all nine variants, including the removal of breast and thyroid, and a magnitude between 0.29 and 0.71. A subgroup effect worth

featuring has to hold under both kinds of perturbation, and only one of the two does.

The site-specific results are internally consistent and match the canon where it is sharp. Lung gives a standardised ERR of 0.666 with a credible female-minus-male ERR difference of 0.480 (0.002, 1.086), against published findings of the largest sex effects for lung with female ERRs three to five times the male values. Colon at 0.297 and liver at 0.291 are lower, as expected. For stomach, banding on stomach dose rather than colon dose gives 0.256 against 0.253 with the same complete absence of heterogeneity, so the flat stomach effect surface is not an artefact of dose banding.

## 1.10 A low-dose city association, and what it turns out to be

### 1.10.1 The association

The overall 5–< 100 mGy contrast is null: the exposed-arm standardised ERR is −0.019 (−0.060, 0.023) and the population-average version is −0.005 (−0.046, 0.037). Nevertheless, city modifies this contrast, and it is the only dimension that does. Of the 49 separation tests run in this fit, five flag on the standardised footing and all five involve city.

Everything in this section is reported on the standardised footing fixed in Section 1.6, and both standardisations are stated once here so that no number below is ambiguous. Over the exposed cells within each city the standardised ERR is −0.043 (−0.089, 0.003) in Hiroshima and +0.088 (0.006, 0.171) in Nagasaki, a difference of 0.132 (0.042, 0.224). Over all cells in each city, which is the footing the separation test uses, the values are −0.041 (−0.087, 0.005) and +0.085 (0.003, 0.167), giving differences from the cohort-wide standardised ATE of −0.036 (−0.063, −0.011) and +0.090 (0.026, 0.159). The two standardisations agree to within 0.005 throughout.

The same contrast computed as a person-year-weighted mean of the per-cell ERRs is about 2.4 times larger, at −0.026 (−0.101, 0.059) in Hiroshima and +0.209 (0.077, 0.365) in Nagasaki. Earlier drafts of this section quoted that figure, and the difference between 0.209 and 0.088 is the whole reason the footing is now fixed in advance: cells with a high ERR are cells with a low baseline rate, so an average of ratios systematically overstates a person-year-weighted rate ratio. The per-cell numbers appear below only where they are labelled as such.

Within Nagasaki the effect concentrates in women, with a standardised ERR of 0.166 (0.054, 0.278), and in those exposed at ages 30–39, 0.165 (0.022, 0.324). It is not a consequence of a high-dose city difference: at 500–< 2000 mGy the two cities are 0.49 (0.37, 0.60) and 0.48 (0.31, 0.67) and do not separate. City is a modifier the canonical model structurally cannot express, since it enters the baseline rate and never the ERR, so this is the clearest instance in the study of the estimator reaching somewhere the established model class cannot.

### 1.10.2 Is it Monte-Carlo noise, and how large is it in dose terms?

Two questions have to be settled before the association is interpreted at all.

The first is Monte-Carlo error, because the lower credible bound on Nagasaki's standardised ERR, 0.0063, excludes zero only barely. Running the contrast as two independent chains and pooling gives, for that quantity, a Gelman–Rubin statistic of 1.002, an effective sample size of 1,468 and a Monte-Carlo standard error of 0.0011, so the lower bound sits 5.9 Monte-Carlo standard errors above zero. On the cohort standardisation the margin is thinner, at 2.6. The Nagasaki-minus-Hiroshima difference is much better resolved, with an effective sample size of 1,926, a Monte-Carlo standard error of 0.0011 and a lower bound 40 Monte-Carlo standard errors above zero. The exclusion of zero is therefore a property of the posterior rather than of the sampler, but Nagasaki's own ERR is only just distinguishable from the null, whereas the difference between the two cities is comfortably so.

The second is magnitude. The low band's mean dose contrast is 0.0299 Gy overall, 0.0315 in Hiroshima and 0.0246 in Nagasaki. Rescaling Nagasaki's standardised ERR by its own dose contrast gives an implied 3.59 (0.26, 6.96) per Gy, and on the per-cell footing 8.49 (3.11, 14.85) per Gy. The cohort's own high-dose slope is 0.529 (0.419, 0.630) per Gy. A dose reading of this result therefore requires Nagasaki's low-dose slope to be roughly seven times the cohort slope on the footing we use, and sixteen times on the footing we do not, while the two cities agree to the second decimal at high dose. That is a strong constraint, and it is what the caveat in Section 1.15 about dividing by 0.03 means in practice.

The E-value sensitivity analysis says the same thing in the other units (Table 14). On the standardised footing an unmeasured confounder with associations of 1.40 with city and with cancer incidence would explain Nagasaki's low-band point estimate away, and 1.09 would move the interval to the null. For comparison, Table 6 gives 1.41 and 1.23 for the medium-dose all-solid result, which Section 1.5 calls less robust, and 2.34 and 2.12 for the high-dose result. The city finding therefore sits at or below the study's own "less robust" benchmark on both measures. Given that the two cities differ in baseline solid-cancer rate by more than 20%, at 66.5 against 54.4 per $10^4$ person-years in the reference arm, a confounder of strength 1.09 is not hard to name.

Table 14: E-values and per-Gy rescaling for the low-dose city contrast, on both footings. Rate ratios are exposed-arm and person-year standardised. The per-Gy column divides by each city's own mean dose contrast and is shown to make the size of the implied dose effect explicit, not because a per-Gy slope is estimated.

| Footing | Quantity | Dose contrast (Gy) | ERR (95% CrI) | Implied ERR per Gy | E-value, point / interval |
|---|---|---|---|---|---|
| Standardised | Hiroshima | 0.0315 | −0.043 (−0.089, 0.003) | −1.38 (−2.81, 0.09) | 1.26 / 1.00 |
| | Nagasaki | 0.0246 | +0.088 (0.006, 0.171) | +3.59 (0.26, 6.96) | 1.40 / 1.09 |
| | Nagasaki − Hiroshima | 0.0299 | +0.132 (0.042, 0.224) | +4.41 (1.40, 7.50) | 1.52 / 1.25 |
| Per-cell mean | Hiroshima | 0.0315 | −0.026 (−0.101, 0.059) | −0.84 (−3.22, 1.86) | 1.19 / 1.00 |
| | Nagasaki | 0.0246 | +0.209 (0.077, 0.365) | +8.49 (3.11, 14.85) | 1.71 / 1.36 |
| | Nagasaki − Hiroshima | 0.0299 | +0.235 (0.100, 0.390) | +7.87 (3.36, 13.06) | 1.77 / 1.43 |
| *Benchmarks from Table 6:* all-solid medium dose 1.41 / 1.23; all-solid high dose 2.34 / 2.12. | | | | | |

### 1.10.3 Is it new relative to the canonical model?

Section 1.7 compares the CountBCF surface with the refitted canonical risk model at the same person-year cells, and its low-dose block answers this directly. Six of the 28 low-dose subgroups depart from the canonical surface on the standardised scale. Hiroshima is one of them, at −0.063 (−0.108, −0.016). Nagasaki is not, at +0.072 (−0.011, 0.154). On the scale this study uses for every other claim, Nagasaki's low-band ERR is not distinguishable from what the canonical model already predicts; Hiroshima's is. All six low-dose departures are downward, so what the estimator is mainly saying about the low band is that the linear extrapolation from the whole 0–2 Gy range overshoots there, and that it overshoots most in Hiroshima. Read that way, the city contrast is at least as much a statement about Hiroshima sitting below the canonical prediction as about Nagasaki sitting above it, which is the first hint that its source is the reference group rather than the exposed group.

### 1.10.4 Is it an artefact of the estimator?

No. The association survives a deliberately broad set of modelling checks (Table 15), covering five moderating-forest tree counts, three moderating-forest depth priors, two multipliers of the leaf

concentration shared by both forests, four propensity specifications, a trim, and the not-in-city group. Across all sixteen the Nagasaki-minus-Hiroshima standardised difference runs from 0.099 to 0.146 and every 95% credible interval excludes zero, including under the tightest prior the sampler will take. As in the high-dose sweep, the magnitude scales with prior looseness while its existence does not: the smallest value, 0.099 (0.028, 0.170), comes from halving the leaf concentration, and the largest, 0.146 (0.049, 0.245), from the most prognostic-like depth prior.

Two labelling points that the earlier draft got wrong are corrected here. The leaf concentration $a_0$ is shared by the prognostic and moderating forests, so those two rows are not a moderating-forest sweep; the tree-count and depth-prior rows are. And the trimming row is now a trim that bites. The propensity score for this contrast spans roughly $[0.50, 0.73]$, so the fixed $[0.10, 0.90]$ trim reported previously removed no cells at all and returned numerically identical results: it was a no-op rather than a passed test. The row now trims to the central 90% of the observed score distribution.

Table 15: Robustness of the low-dose (5–< 100 mGy) city difference to modelling choices. Each row is a separate refit on the standardised footing. $m_\tau$ and the depth prior act on the moderating forest only; the leaf concentration $a_0$ is shared by both forests. The Nagasaki-minus-Hiroshima difference excludes zero in all sixteen specifications, and also on the per-cell footing, which is in the accompanying CSV. These fits use the shorter sensitivity MCMC budget of one chain with 1,000 burn-in and 1,000 retained draws, so the default row reads 0.128 against the 0.132 obtained from the two-chain headline fit.

| Specification | Cells | Hiroshima std. ERR | Nagasaki std. ERR | Nagasaki − Hiroshima (95% C |
|---|---|---|---|---|
| $m_\tau = 10$ (tightest) | 8,541 | −0.033 | 0.079 | 0.112 (0.025, 0.200) |
| $m_\tau = 25$ | 8,541 | −0.034 | 0.082 | 0.116 (0.026, 0.207) |
| $m_\tau = 50$ (default) | 8,541 | −0.035 | 0.093 | 0.128 (0.038, 0.217) |
| $m_\tau = 100$ | 8,541 | −0.042 | 0.101 | 0.143 (0.051, 0.238) |
| $m_\tau = 200$ (loosest) | 8,541 | −0.040 | 0.103 | 0.142 (0.050, 0.241) |
| Depth prior (0.05, 4), tight | 8,541 | −0.040 | 0.083 | 0.123 (0.036, 0.217) |
| Depth prior (0.5, 2), loose | 8,541 | −0.042 | 0.102 | 0.144 (0.051, 0.243) |
| Depth prior (0.95, 2), prognostic-like | 8,541 | −0.043 | 0.103 | 0.146 (0.049, 0.245) |
| $a_0 \times 0.5$ (both forests) | 8,541 | −0.047 | 0.052 | 0.099 (0.028, 0.170) |
| $a_0 \times 2$ (both forests) | 8,541 | −0.044 | 0.088 | 0.132 (0.039, 0.220) |
| Propensity score constant | 8,541 | −0.038 | 0.089 | 0.127 (0.034, 0.224) |
| Propensity score simple logistic | 8,541 | −0.040 | 0.096 | 0.136 (0.044, 0.228) |
| Propensity score rich logistic | 8,541 | −0.038 | 0.087 | 0.126 (0.040, 0.222) |
| Propensity score random forest | 8,541 | −0.028 | 0.094 | 0.122 (0.022, 0.227) |
| Trimmed to the central 90% of $\hat{\pi}$ | 7,687 | −0.047 | 0.088 | 0.135 (0.046, 0.224) |
| Not in city added to control arm | 9,826 | −0.039 | 0.103 | 0.142 (0.050, 0.239) |

An independently fitted adjusted Poisson model with dose-band by city interactions gives the same localisation. The interaction term improves the fit at $p = 0.012$ on a likelihood-ratio test with three degrees of freedom, and the improvement is carried entirely by the low band (Table 16), where the rate-ratio ratio is 1.154 (1.054, 1.264). The finding is therefore not a CountBCF artefact but a feature of the data that a conventional model also sees once it is told to look for it, and it has since been reproduced independently in Poisson regression by a co-author at RERF. What CountBCF adds is that it located the modifier without being told. It should be said plainly, though, what this parametric check does and does not establish. It was run on the same data and the same variable

after the forest result was known, so it establishes that the association is in the data rather than in the estimator. It is not independent evidence against multiplicity, and it does not address any of the explanations below, because it shares the same reference group.

Table 16: Dose-band by city interaction terms from an adjusted Poisson model. The interaction is confined to the low band. Entries are rate-ratio ratios relative to Hiroshima with 95% confidence intervals.

| Interaction term | Rate-ratio ratio (95% CI) | $p$ |
|---|---|---|
| 5–< 100 mGy × Nagasaki | 1.154 (1.054, 1.264) | 0.002 |
| 100–< 500 mGy × Nagasaki | 1.111 (0.976, 1.263) | 0.11 |
| 500–< 2000 mGy × Nagasaki | 1.001 (0.870, 1.151) | 0.99 |

### 1.10.5 Reconciliation with the published city analyses

The canonical 1958–1998 analysis reported a significant low-dose dose response but did not foreground a city-specific split [Preston et al., 2007], and the 1958–2009 update found little evidence that city modified the aggregate solid-cancer response, estimating Nagasaki's radiation effect at about 12% below Hiroshima's with an interval covering no difference [Grant et al., 2017]. Those analyses are not in conflict with what we report. They evaluate city differences in the magnitude and shape of the dose response across the full dose range, and they were motivated principally by dosimetry, in particular the larger neutron contribution in Hiroshima. A full-range comparison is dominated by the high-dose contribution, where we also find no city difference at all. What is genuinely under-examined in the published literature is a city comparison confined to the lowest band, and that is the gap this section occupies.

### 1.10.6 Explanation 1: the composition of the zero-dose reference group

This is the explanation the data support, and it is a known problem with a known name. French et al. [2017] show that in the LSS the choice of reference group among zero-dose survivors materially changes the estimated ERR and the inferred curvature, especially below 1 Gy, and they recommend adjusting for city-specific risk differences among distal survivors rather than pooling them. Cologne et al. [2019] show that heterogeneity in background incidence, including city differences in stomach cancer, *Helicobacter pylori* prevalence and smoking, distorts inference about the aggregate solid-cancer dose response. Neither of those mechanisms is representable by any covariate in the public file.

The geography is the reason the reference group is not exchangeable across cities. Hiroshima's hypocentre was directly above the downtown area, and because the surrounding terrain is relatively flat the damage pattern was largely radial; moving outward from the hypocentre the population becomes progressively more rural in every direction, so Hiroshima's distal survivors beyond 3 km were largely rural at the time of the bombing. Nagasaki's hypocentre was north of downtown, in the Urakami district, and the Urakami valley is enclosed by mountain ranges; most of Nagasaki's distal survivors, 3 km and beyond to the south and southeast, were urban residents. The same nominal stratum, "zero-dose survivor beyond 3 km", therefore denotes a rural population in one city and an urban one in the other. A further asymmetry comes from cohort construction: after the initial selection of LSS participants, additional distal survivors were added to the Nagasaki cohort, randomly selected to match the original Nagasaki cohort on sex and age, which raises both the number and the proportion of distal survivors in Nagasaki. We note that "distal" is conventionally

$\geq$ 3 km although the original selection used $\geq$ 2.5 km, because dose estimates were not available when the cohort was assembled and distance was the best available surrogate for separating exposed from virtually unexposed survivors.

Our own reference arm has exactly this structure, and the imbalance is large (Table 17). In Hiroshima 73.9% of reference-arm person-years come from beyond 3 km; in Nagasaki, 55.7%. Meanwhile the distal ring carries a higher adjusted solid-cancer rate than the outer proximal ring in both cities, which is the direction French et al. [2017] report. Pooling the two rings therefore raises the reference rate more in Hiroshima than in Nagasaki, which depresses Hiroshima's ERR and inflates Nagasaki's – which is the sign and, to within the precision available, the size of the observed city contrast.

Table 17: Composition of the 0–< 5 mGy reference arm by city and distance ring, with the 5–< 100 mGy band for comparison. Crude rates are cases per $10^4$ person-years; the adjusted rates come from a Poisson model with the same sex-specific spline baseline used throughout. The exposed band contains no cell beyond 2.71 km, so the distal ring has probability zero of low-band membership.

| City | Stratum | Ground distance (km) | Subjects | Person-years | Crude rate | Adjusted rate |
|---|---|---|---|---|---|---|
| Hiroshima | Reference, outer proximal | 2.47–3.00 | 5,197 | 139,394 | 64.7 | 60.9 |
| | Reference, distal | 3.32–7.58 | 14,842 | 394,595 | 67.0 | 64.2 |
| | Low band, 5–< 100 mGy | 1.60–2.51 | 21,105 | 566,405 | 61.5 | — |
| Nagasaki | Reference, outer proximal | 2.52–2.97 | 6,836 | 170,049 | 52.1 | 56.5 |
| | Reference, distal | 3.22–6.97 | 8,670 | 214,161 | 54.8 | 58.2 |
| | Low band, 5–< 100 mGy | 1.63–2.71 | 6,684 | 163,199 | 56.4 | — |

The arithmetic is worth doing explicitly, because it predicts the size of the effect and not only its direction. Weighting the two adjusted ring rates by each city's own person-year shares gives a pooled reference rate of 63.3 in Hiroshima against 60.9 for the outer proximal ring alone, a ratio of 1.040, and 57.5 against 56.5 in Nagasaki, a ratio of 1.017. Moving from the pooled reference to the outer proximal reference should therefore move the two cities' ERRs apart by roughly $1.040/1.017-1 = 0.023$ in Hiroshima's favour, that is, shrink the Nagasaki-minus-Hiroshima difference by about 0.023. The observed shrinkage is 0.029.

Table 18 reports what actually happens. Two constructions are the ones the literature recommends: restricting the reference arm to the outer proximal ring, which is the RERF practice, and keeping the full reference arm but giving the prognostic forest a distal indicator, which is the French et al. [2017] recommendation of adjusting for city-specific risk differences among distal survivors. Under both, the Nagasaki-minus-Hiroshima difference falls from 0.134 (0.041, 0.231) to 0.105 (−0.018, 0.222) and 0.106 (−0.011, 0.223) respectively, and in both the interval covers zero. The same is true on the per-cell footing, where the difference falls from 0.225 (0.090, 0.388) to 0.154 (−0.020, 0.324) and 0.145 (−0.037, 0.322).

Two qualifications on that result. First, the effect is a 22% attenuation of the point estimate combined with a wider interval, and the widening is partly mechanical: the outer-proximal reference arm carries 6,194 cases against 10,009, so precision is lost as well as signal. What the exercise establishes is not that the difference is zero but that it is not robust to the reference-group choice known to matter most for exactly this question. Second, adding the ring to the propensity score rather than to the prognostic forest attenuates less, to 0.122 (0.009, 0.239) and 0.118 (0.015, 0.233), and those intervals stay marginally clear of zero; but those are also the specifications in which the fitted score runs down to $5 \times 10^{-9}$, because the distal reference cells have probability exactly zero of low-band membership. Conditioning on the ring in the score is the wrong instrument for that

reason, which is precisely why the recommended constructions handle the distal group through the baseline instead.

The last row of the table is the check Igor asked to see reported in full. Putting continuous ground distance into both the score and the prognostic forest reverses the sign of the city difference, to −0.084 (−0.332, 0.156), and destroys positivity completely: the fitted propensity spans $2 \times 10^{-16}$ to 1.000. That fit is uninterpretable as a causal contrast and is reported only to document the sign reversal, which was previously mentioned in Section 1.2 and nowhere else.

Table 18: The low-dose city difference under alternative treatments of the $< 3/\geq 3$ km distance ring. All entries are on the standardised footing over exposed cells; the per-cell column is given for cross-reference. The first two rows after the baseline are the constructions recommended by RERF practice and by French et al. [2017]. The last row conditions on the continuous assignment variable and is reported only to document the sign reversal and the positivity collapse.

| Specification | Cases | $\hat{\pi}$ range | Nagasaki − Hiroshima, std. (95% CrI) | Same, per-cell (95% CrI) | Separates? |
|---|---|---|---|---|---|
| Baseline, full 0–< 5 mGy reference | 10,009 | 0.50–0.73 | 0.134 (0.041, 0.231) | 0.225 (0.090, 0.388) | Yes |
| Reference = outer proximal only (2.5–3.0 km) | 6,194 | 0.69–0.84 | 0.105 (−0.018, 0.222) | 0.154 (−0.020, 0.324) | No |
| Distal indicator in the prognostic forest | 10,009 | 0.50–0.73 | 0.106 (−0.011, 0.223) | 0.145 (−0.037, 0.322) | No |
| Reference = distal only (3–10 km) | 8,221 | 0.61–0.85 | 0.103 (0.014, 0.196) | 0.170 (0.049, 0.308) | Yes |
| City × distance in the score only | 10,009 | $5 \times 10^{-9}$–0.84 | 0.118 (0.015, 0.233) | 0.154 (−0.008, 0.324) | Yes |
| City × distance in score and forest | 10,009 | $5 \times 10^{-9}$–0.84 | 0.122 (0.009, 0.239) | 0.182 (−0.024, 0.374) | Yes |
| Continuous ground distance in score and forest | 10,009 | $2 \times 10^{-16}$–1.000 | −0.084 (−0.332, 0.156) | −0.063 (−0.289, 0.151) | No |

The parametric model reaches the same conclusion, which matters because it is free of any question about the forest and because it is the form a radiation epidemiologist would fit (Table 19). Refitting the dose-band by city interaction of Table 16 under the two recommended reference constructions moves the joint likelihood-ratio test from $p = 0.012$ to $p = 0.106$ and $p = 0.108$, and the low-band interaction from 1.154 (1.054, 1.264) at $p = 0.002$ to 1.131 (1.006, 1.273) at $p = 0.040$ and 1.131 (1.005, 1.272) at $p = 0.041$. Restricting the reference to the distal ring instead, which is the construction that should not help if the mechanism is what we claim, leaves the result essentially intact at $p = 0.029$.

Table 19: The parametric dose-band by city interaction under four reference-group constructions. Entries are rate-ratio ratios relative to Hiroshima with 95% confidence intervals; the final column is the joint likelihood-ratio test on three degrees of freedom for the whole interaction.

| Reference construction | 5–< 100 mGy × Nagasaki | $p$ | Joint LRT $p$ |
|---|---|---|---|
| Primary, full 0–< 5 mGy reference | 1.154 (1.054, 1.264) | 0.002 | 0.012 |
| City-specific distal baseline term | 1.132 (1.006, 1.273) | 0.040 | 0.106 |
| Reference = outer proximal only | 1.131 (1.005, 1.272) | 0.041 | 0.108 |
| Reference = distal only | 1.151 (1.042, 1.272) | 0.006 | 0.029 |

### 1.10.7 Explanation 2: a dose effect must have a gradient; this does not

The reference-group explanation and a dose explanation make opposite predictions inside the band, and the prediction is easy to test. A radiation effect must grow with dose: the ERR per Gy should be roughly the same in each sub-band, so the ERR itself should rise as the sub-band mean dose rises. A population offset attached to being in the exposed ring at all makes no such prediction: it produces a step at the 5 mGy boundary and then a flat ERR across the rest of the band, so the implied ERR per Gy collapses as the divisor grows.

Splitting the low band into 5–< 20, 20–< 50 and 50–< 100 mGy and refitting each against the same 0–< 5 mGy reference gives the second pattern, not the first (Table 20). Nagasaki's standardised ERR is 0.118 (0.010, 0.226) at 5–< 20 mGy, 0.044 (−0.077, 0.166) at 20–< 50 mGy and 0.101 (−0.067, 0.287) at 50–< 100 mGy: no rise with dose. The Nagasaki-minus-Hiroshima difference is 0.167 (0.049, 0.282), 0.111 (−0.024, 0.253) and 0.110 (−0.069, 0.298): a level, not a gradient, and credible only in the sub-band where there is essentially no dose at all. The implied ERR per Gy in Nagasaki accordingly falls from 13.0 at 5–< 20 mGy, where the dose contrast is 9.1 mGy, to 1.28 and 1.37 in the two higher sub-bands. A tenfold collapse in the apparent per-Gy slope as one moves from 9 mGy to 73 mGy of contrast is the signature of a fixed offset divided by a growing denominator.

Two honest qualifications. The exposed arm shrinks as the sub-band narrows, from about 2,080 exposed cases at 5–< 20 mGy to about 880 at 50–< 100 mGy, so some of the loss of credibility in the upper sub-bands is power rather than signal. And the point estimate of the city gap does not vanish above 20 mGy; it stays near 0.11. But that is precisely the point: a gap that stays at roughly the same *level* while the dose contrast grows eightfold is not a dose response, whatever else it is.

Table 20: The low band split by dose, each sub-band contrasted against the same 0–< 5 mGy reference. A dose effect predicts a roughly constant ERR per Gy down the last two columns; a population offset predicts a roughly constant ERR in the middle columns and a collapsing ERR per Gy. The data show the second.

| Sub-band | Dose contrast (Gy) | Hiroshima std. ERR | Nagasaki std. ERR | Nagasaki − Hiroshima | Sep.? | Nagasaki ERR/Gy |
|---|---|---|---|---|---|---|
| 5–< 20 mGy | 0.0096 | −0.049 (−0.109, 0.011) | +0.118 (0.010, 0.226) | 0.167 (0.049, 0.282) | Yes | 13.0 |
| 20–< 50 mGy | 0.0353 | −0.068 (−0.129, −0.002) | +0.044 (−0.077, 0.166) | 0.111 (−0.024, 0.253) | No | 1.28 |
| 50–< 100 mGy | 0.0743 | −0.010 (−0.087, 0.069) | +0.101 (−0.067, 0.287) | 0.110 (−0.069, 0.298) | No | 1.37 |

### 1.10.8 Explanations the data address less well

Cancer-site composition is not the driver. Repeating the low-band contrast on the aggregate with breast and thyroid removed, with non-melanoma skin removed, with stomach removed, and restricted to non-sex-specific sites gives a Nagasaki-minus-Hiroshima difference of 0.129 (0.031, 0.228), 0.124 (0.032, 0.222), 0.142 (0.034, 0.251) and 0.107 (0.016, 0.206) respectively, against the baseline 0.134 (0.041, 0.231); all four still separate. The result is therefore not an artefact of one or two sites with city-specific background structure, in the way that Sasaki et al. [2014] attribute a low-dose Nagasaki excess mainly to lung, liver and gallbladder. It is a property of the aggregate, which is what a reference-group explanation predicts and a site-specific one does not.

Dose assignment remains a live possibility that we cannot settle. The two cities have different shielding histories and different gamma-to-neutron mixtures, so an error in the neutron weighting or in the distal dose imputation would present as a city difference in risk per weighted Gy. Two considerations cut against it being the main explanation: the neutron fraction is smallest at exactly these doses, and the cities agree at high dose where the neutron component is largest, which is the opposite of what a neutron-weighting error predicts. Residual and fallout dose is a different matter. It is not in DS02 at all, it is city-specific, and if present it would make the result a genuine radiation effect misattributed to city. Nothing in the public file can address it.

Case ascertainment is the explanation we can say least about from the data and most about from domain knowledge. LSS incidence, unlike mortality, depends on the two tumour registry catchment areas, and person-years are adjusted for migration into and out of them [Grant et al., 2017]; that

adjustment is city-specific and varies with distance from the city centre, which is the axis of this finding. We are advised by our RERF co-author that migration adjustment, case ascertainment, registry quality, Adult Health Study participation and examination protocols are comparable between the two cities. The one relevant asymmetry is the Nagasaki distal augmentation described in Section 1.10.6, which acts on the reference arm along precisely the axis identified there. The decisive check would be to repeat the low-band city contrast on the LSS mortality file, which is free of the catchment and migration problem; that file is not in the public release, and the check is recorded in Section 1.15 as the first item for follow-up.

Multiplicity cannot be dismissed either. This is one flagged result among 294 separation tests (Table 7), and $p = 0.012$ from the parametric interaction is not overwhelming, particularly since that test was run on the same data and the same variable after the forest result was known.

### 1.10.9 What we conclude

The low-dose city contrast is a real feature of the data and is not an artefact of CountBCF: it survives every prior, propensity and trimming perturbation, and a conventional Poisson model sees it too. It is not, on this evidence, a radiation finding. On the standardised footing Nagasaki's low-band ERR is not distinguishable from the canonical model's own prediction; its E-values sit at or below the study's "less robust" benchmark; a dose reading would require a low-dose slope several times the cohort slope while the two cities agree exactly at high dose; the difference is attenuated and loses credibility under the two reference-group constructions the LSS literature recommends for exactly this problem; and it does not show the dose gradient inside the band that a dose effect must produce. What the data support is that the zero-dose reference group is not exchangeable across the two cities, for reasons of geography and cohort construction that are documented in the LSS literature and that no covariate in the public file can represent.

That is the result we report, and it is a better demonstration of the method than a novel city claim would have been. CountBCF was told nothing about city except to let it modify the effect, and it identified the one place where the canonical model class is structurally unable to express what the data contain, because city enters the baseline rate and never the ERR. Following that flag with the standard epidemiological tools then located the structure in the reference group. The division of labour is the point: the estimator says where to look, and the domain analysis says what is there.

## 1.11 Propensity score and overlap

The primary propensity score is a logistic regression of band membership on city by sex, natural splines with four degrees of freedom in age at exposure and in attained age, and a city by age-at-exposure interaction. Ground distance is deliberately excluded because it is the assignment mechanism rather than a confounder. Four alternatives were run for comparison: a constant score, a plain additive logistic model, a saturated spline model, and a probability random forest.

Overlap is not strained anywhere. For the featured high-dose contrast the primary score spans 0.117 to 0.752, with no cells below 0.05 or above 0.95, and the same holds for the low and medium contrasts and for every logistic specification. Only the random forest, which overfits with an area under the curve of 0.77 against 0.56 for the logistic models, pushes two cells above 0.95. The score is nearly flat because dose was determined by location and shielding rather than by demography, so the discrimination of the logistic model is barely above chance. Figure 8 displays the distributions.

Weighting removes what imbalance is present. Cell-level standardised differences before weighting run from 0.056 to 0.110 for city and reach $-0.206$ for age at exposure; after inverse-probability

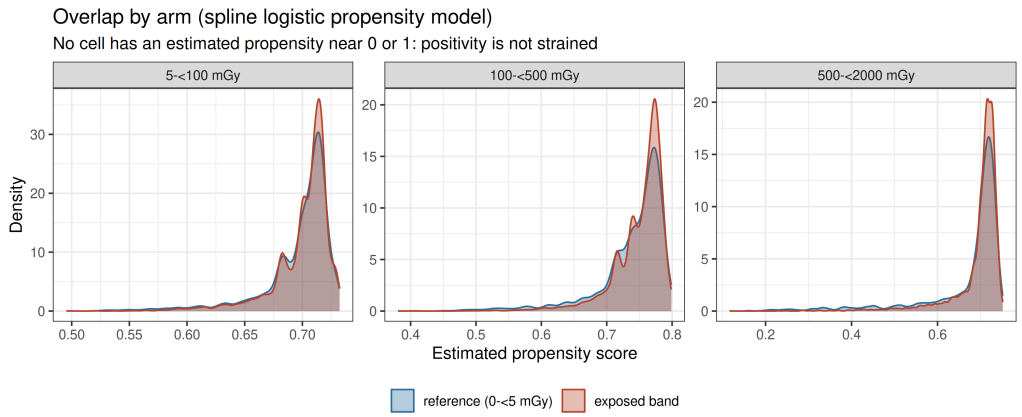



Figure 8: Propensity-score distributions by arm for each dose-band contrast. No cell falls below 0.05 or above 0.95 under the primary specification.

weighting with the primary score they are −0.003 to 0.002 for city, −0.002 for age at exposure, −0.003 for attained age, and −0.002 for sex. Trimming is a no-op: because no cell falls outside $[0.10, 0.90]$, trimming at either $[0.05, 0.95]$ or $[0.10, 0.90]$ removes nothing and returns numerically identical results, with all three variants giving 8,249 cells, a standardised ERR of 0.495, and an age-at-exposure ERR difference of 0.795. Capping and overlap weights are therefore unnecessary here.

Table 21: Sensitivity of the featured contrast to the propensity-score specification. Even a constant score reproduces the primary estimate, because the prognostic forest already conditions on city, sex, and both age variables.

| Propensity specification | Cells | Standardised ERR | EAR/$10^4$ PY (95% CrI) | ERR, $<20$ minus 40+ |
|---|---|---|---|---|
| Constant (no covariates) | 8,249 | 0.497 | 30.8 (25.1, 36.7) | 0.815 (0.473, 1.19) |
| Simple logistic | 8,249 | 0.500 | 30.9 (25.1, 36.9) | 0.844 (0.494, 1.21) |
| Spline logistic (primary) | 8,249 | 0.495 | 30.7 (25.3, 36.5) | 0.795 (0.447, 1.14) |
| Saturated spline logistic | 8,249 | 0.491 | 30.6 (25.4, 36.2) | 0.773 (0.429, 1.13) |
| Probability random forest | 8,249 | 0.497 | 30.8 (25.3, 36.4) | 0.913 (0.464, 1.39) |

The answer does not depend on the score at all (Table 21). Even a constant propensity gives 0.497 against the primary 0.495, because the prognostic forest is already conditioning on city, sex, and both ages, leaving the score little to correct. That is a reassuring finding rather than a disappointing one: the identification here does not hinge on getting the score right.

The featured subgroups are also not in a low-overlap tail. Within the exposed arm of the high-dose contrast, the youngest-at-exposure cells, which carry the large ERR, have the tightest propensity distributions of all, spanning 0.700 to 0.731 for males exposed at ages 0–9 and 0.714 to 0.742 for females, with no cell below 0.05 or above 0.95. The only subgroup with any low-propensity cells is those exposed at age 40 or later, with a minimum of 0.183, and that subgroup sits at the low end of the ERR gradient. The main surface is therefore not an artefact of a low-overlap tail.

One limitation of these diagnostics has to be stated explicitly, because it is the hinge of Section 1.10. They are computed from a propensity model that deliberately omits ground distance, the assignment variable. What they establish is that the arms overlap on the demographic covariates, which they do, comfortably. What they cannot establish is overlap on anything distance-correlated,

because within a city the dose band is very nearly a distance ring: the 5–< 100 mGy band runs from about 1.6 to 2.7 km from the hypocentre and the reference arm from 2.5 km outward, with no demographic variable in the file distinguishing the two. A flat propensity score and clean balance tables are consistent with a large unadjusted difference in the urban and rural composition of the two rings, and, as Section 1.10 shows, that is what the data contain. The sharpest statement of the problem is that the 5–< 100 mGy band contains no cell beyond 2.71 km, while 74% of Hiroshima's reference person-years and 56% of Nagasaki's come from beyond 3.22 km. Those distal reference cells have a probability of low-band membership of exactly zero, which is the same positivity argument that Section 1.2 uses to exclude the not-in-city group, applied one ring closer in. Adding the ring to the propensity model makes this visible immediately: the fitted score then runs from essentially zero rather than from 0.50.

### 1.12 Repeated person-years

Each survivor contributes person-years to many cells as they move through attained-age and calendar-period categories, so an independent-cell likelihood should in principle understate the variance. We quantify this with a frequentist cluster bootstrap of the parametric model, resampling the 1,128 survivor cohorts, defined by city, sex, age at exposure, dose category, and distance category, with replacement over 300 replicates (Table 22).

Table 22: Cluster bootstrap of the parametric model over 300 replicates of the 1,128 survivor cohorts. The design effect is the ratio of the bootstrap sampling variance to the model variance (the square of the ratio of the bootstrap standard error to the model standard error).

| Dose band | Coefficient | Model SE | Cluster-bootstrap SE | Design effect |
|---|---|---|---|---|
| Low | −0.0036 | 0.0205 | 0.0211 | 1.06 |
| Medium | 0.1112 | 0.0259 | 0.0247 | 0.91 |
| High | 0.4261 | 0.0325 | 0.0346 | 1.14 |

The clustering penalty is at most about 7% on the standard error, and the point estimates do not move. The reason is structural: each cell already aggregates hundreds of survivors, and the exposure is constant within a survivor cohort, so the between-cell correlation induced by repeated person-years is small relative to the Poisson sampling noise in cells that are 74% zeros. This bounds the precision of the independent-cell likelihood without changing any conclusion [Field and Welsh, 2007, Kish, 1995].

We do not report a CountBCF random-intercept fit alongside the bootstrap. The current implementation accepts a random-effects design and samples the coefficients, but they never enter the count likelihood: the fitted log mean is the offset plus the prognostic term plus the treatment term only. The switch is therefore inert, and this has a consequence worth stating plainly, because it is easy to be misled by it. Any fit made with the random-effects arguments enabled is numerically an independent-cell fit that additionally reports a sampled variance component; the variance component is a prior draw and carries no information about the survivor-cohort structure. Two rows of our own sensitivity output were originally labelled "random intercepts" on that basis and were not random-intercept fits. They have been replaced by the mechanism the package documents for group structure, a survivor-cohort indicator supplied to the prognostic forest, which does enter the likelihood. That substitution is a real test and it passes: giving the prognostic forest a 60-level city by sex by age-at-exposure indicator moves the featured standardised ERR from 0.495 (0.401, 0.594) to 0.494 (0.399, 0.592), and giving it the 432-level finest survivor cohort moves it to 0.507

(0.403, 0.616). The credible interval widens by 3% and 6% respectively, which brackets the 7% the cluster bootstrap reports and is the same answer by a different route. Adding a random-effects component that enters the count likelihood is on the development list for the method.

### 1.13 Low-dose sub-cohorts

Dropkin [2016] re-analysed this same file with generalized additive models restricted to women and to the 0–20, 0–100, and 0–500 mGy sub-cohorts, and reported relative risks at 10 mGy that are ten to forty-five times higher than a linear extrapolation from the full cohort. The same restrictions were run here (Table 23).

Table 23: CountBCF fits on low-dose sub-cohorts. The dose contrast column is the mean weighted-dose separation between the exposed and reference groups.

| Sub-cohort | Cells | Cases | Dose contrast (Gy) | Standardised ERR (95% CrI) | EAR/$10^4$ PY (95% CrI) |
|---|---|---|---|---|---|
| 5–< 100 mGy, women | 4,360 | 5,474 | 0.029 | 0.0006 (−0.061, 0.059) | 0.01 (−3.43, 3.13) |
| 5–< 100 mGy, men | 4,181 | 4,535 | 0.031 | −0.047 (−0.110, 0.012) | −3.60 (−8.61, 0.90) |
| 5–< 20 mGy, both sexes | 3,839 | 7,687 | 0.0096 | −0.005 (−0.057, 0.049) | −0.28 (−3.36, 2.90) |
| 5–< 20 mGy, women | 1,938 | 4,239 | 0.0096 | −0.004 (−0.075, 0.065) | −0.25 (−4.09, 3.47) |

CountBCF finds no low-dose excess in any of them. This is not a refutation of Dropkin [2016] and should not be presented as one, because the two designs answer different questions. His model fits a smooth dose response in a transformed dose coordinate, the empirical cumulative distribution function of dose, and reads a risk off it at 10 mGy, borrowing strength across the whole low-dose range. Ours contrasts a 5–< 20 mGy band against a 0–< 5 mGy band, a mean dose separation of only 9.6 mGy, with no functional form to borrow from. At that separation the design has essentially no power: the credible interval on the ERR, at roughly ±0.07, corresponds to ±7 per Gy, which cannot discriminate a null from any of the low-dose hypotheses in play. The honest statement is that the categorical-contrast design, adopted for clarity of the causal estimand, is the wrong instrument for the low-dose question, and that a continuous-dose extension of CountBCF is what would be needed to engage with that result directly.

### 1.14 Excess cases and attributable fractions

Converting the exposed-arm effect back to a case count gives the comparison in Table 24. At medium and high dose the two approaches agree closely, at 372 against 402 excess solid cancers in the 0.5–2 Gy range and 178 against 254 in the 0.1–0.5 Gy range. The gap widens at low dose, where CountBCF returns a null with a slightly negative point estimate while the parametric model assigns 81 excess cases. That is the linear dose-response assumption doing work: the parametric model borrows the slope from the whole 0–2 Gy range and applies it at 30 mGy, whereas the band contrast must find the signal within the band itself. Neither is wrong, but they answer different questions, and the difference illustrates concretely what a nonparametric causal estimator will and will not deliver.

### 1.15 Caveats and open items

Several limitations qualify the results above and are stated here in full.

The low-dose city difference is robust to modelling and not robust to the reference group. It survives every prior, propensity and trimming perturbation we applied and is independently visible in a parametric interaction model, so it is not an artefact of the estimator. It does not survive

Table 24: Excess cases and attributable fractions from CountBCF, compared with the corresponding published estimates. AF denotes the attributable fraction among the exposed.

| Endpoint | Band | Observed cases | CountBCF excess (95% CrI) | CountBCF AF | Published excess | Published AF |
|---|---|---|---|---|---|---|
| All solid | 5–< 100 mGy | 4,406 | −86 (−274, 103) | −2.0% | 81 | 1.8% |
| | 100–< 500 mGy | 2,112 | 178 (68, 286) | 8.4% | 254 | 12.0% |
| | 500–< 2000 mGy | 1,148 | 372 (300, 438) | 32.4% | 402 | 35.0% |
| Stomach | 5–< 100 mGy | 1,230 | −58 (−165, 48) | −4.7% | 14 | 1.2% |
| | 100–< 500 mGy | 555 | 0.3 (−57, 60) | 0.0% | 46 | 8.2% |
| | 500–< 2000 mGy | 269 | 54 (21, 89) | 20.0% | 69 | 25.9% |

the reference-group construction that RERF uses for exactly this problem, and it does not behave like a dose effect when the band is split by dose. Its E-values sit at or below this study's own "less robust" benchmark. It should be reported as what it is: a real feature of the data that the domain analysis attributes to the composition of the zero-dose reference group, not evidence of a city difference in radiosensitivity. Two explanations that we cannot rule out with the public file remain open, namely residual and fallout dose, which is city-specific and absent from DS02, and multiplicity.

The high-dose sex effect should be reported as prior-sensitive in magnitude and as carried by breast and thyroid. On the standardised footing its credible interval excludes zero under all twelve regularisation settings, so its existence is prior-robust, but its size falls ninefold across that sweep, and it does not survive the data perturbations: its interval covers zero once breast and thyroid are removed, once stomach is removed, and in the non-sex-specific fit. Its agreement with the published female-to-male ERR ratio is nevertheless close on the standardised footing, at 1.54 against 1.6. The medium-dose sex effect does not separate at all on the standardised footing, separates marginally on the per-cell footing, and has not been put through the regularisation sweep; it is not a finding.

The band-contrast design is deliberately blunt at low dose. The 5–< 100 mGy contrast has a mean dose separation of 30 mGy, so nothing about the broader low-dose controversy can be resolved with it. This cuts both ways, and it is the single most important quantitative caveat on Section 1.10: any ERR expressed per Gy in that band is divided by a number close to 0.03, so a small difference in baseline composition between two groups is magnified roughly thirtyfold when it is read as a dose effect.

CountBCF's flat stomach effect surface is a defensible consequence of differential shrinkage on a weak signal, and it is stable across dose banding. The twelve-setting sweep in Table 12 now tests the regularisation concern directly: all eleven non-extreme rows, including the default repeats, produce no subgroup separation on either ERR footing, with a small surface standard deviation. The lone exception is the extreme shared $a_0 \times 2$ setting, which produces an unstable surface with 9 of 49 standardised flags and 15 of 49 per-cell flags. That pathological fit is not evidence for a stomach modifier; the ordinary sweep and the published significance tests agree that the stomach data do not support one.

Every unit of analysis is a cell rather than a person, so all weighting is by person-years and no individual-level covariate such as smoking, diet, or *Helicobacter pylori* status is available. The stomach results in particular are confounded by unmeasured causes known to matter a great deal for that site in Japan.

The public table carries no information about case ascertainment. LSS incidence, unlike mortality, depends on the two tumour registry catchment areas, and the published person-years are adjusted for migration into and out of them [Grant et al., 2017]. That adjustment is city-specific and varies with distance from the city centre, which is the same axis as the low-dose city contrast. We

are advised by the RERF co-author on this paper that migration adjustment, case ascertainment, registry quality, the proportion of Adult Health Study participants, and examination protocols are comparable between Hiroshima and Nagasaki, with one exception: after the initial selection of LSS participants, additional distal survivors were later added to the Nagasaki cohort, randomly selected to match the original Nagasaki cohort on sex and age. That augmentation raises both the number and the proportion of distal survivors in Nagasaki, and it therefore acts on the reference arm along exactly the axis Section 1.10 identifies. None of this is representable in the public file.

Ground distance is excluded from the primary propensity model and from both forests. That exclusion is correct, since distance is the dose-assignment mechanism rather than a pre-exposure confounder, but within a city the dose band is nearly the same object as a distance ring. Anything distance-correlated and city-specific is therefore unadjustable by construction in the primary specification, and the overlap diagnostics of Section 1.11 do not speak to it, because they come from a score that omits the assignment variable. The additional fits reported in Section 1.10, which do use the coarse $< 3/\geq 3$ km ring, are the study's answer to this, and they change the low-dose city result.

The dose is a person-year-weighted cell mean of an adjusted, truncated DS02 organ dose with an assumed 35% dose error already corrected in the RERF release. Dose-measurement error beyond that correction is not modelled. Residual and fallout dose is not in DS02 at all and is city-specific, so a component of it would appear in our analysis as a city difference rather than as a dose effect.

Finally, one check we would have liked to run is not available in the public data. Replicating the low-band city contrast on the LSS mortality file would be free of the registry catchment and migration problem entirely, and would separate an ascertainment explanation from a baseline-rate explanation. That file is held by RERF and the check is left as the first item for a follow-up analysis.

### 1.16 Summary

On the full LSS incidence table, CountBCF reproduces the established average solid-cancer dose response and the main age-at-exposure and attained-age gradients without imposing their functional form. Its agreement with the refitted canonical surface is close over most of the observed covariate space, agreeing to four significant figures on the person-year-weighted mean excess rate at high dose, while localized departures at young attained age and short latency identify regions in which parametric extrapolation deserves scrutiny. The age-at-exposure gradient survives every regularisation setting and every data perturbation, whereas the high-dose sex effect survives the first but not the second, being carried by breast and thyroid, and its magnitude varies ninefold across the regularisation sweep even where its interval stays clear of zero, although on the standardised footing it agrees closely with the published female-to-male ratio. The model appropriately declines to create stomach-cancer heterogeneity unsupported by the site-specific data. Demographic positivity is not strained, the answer does not depend on the propensity specification, and the clustering induced by repeated person-years costs at most about 7% on the standard error. Positivity on the distance axis is a different matter, and it is the study's main identification limitation: ground distance is the dose-assignment mechanism and is therefore excluded from the score and the forests, but within a city the dose band is nearly a distance ring, so conditioning on distance collapses overlap and reverses the sign of the estimate, and anything distance-correlated and city-specific is unadjustable by construction in the primary specification.

The low-dose city contrast is best read as a demonstration of what the method is for rather than as a new radiation finding. CountBCF located, without being told to look, a modifier that the canonical model class structurally cannot express, since city enters the baseline rate and never the ERR. The association is robust to every modelling choice we varied and is independently visible in

a conventional Poisson interaction model. It is not robust to the reference-group construction that RERF uses for precisely this problem, and it does not behave like a dose effect when the band is split by dose. What the forest found is real structure in the data, and the domain follow-up locates that structure in the composition of the zero-dose reference group rather than in radiation. That is the intended division of labour: the estimator identifies where the canonical model is being asked to do something it cannot, and the epidemiology then says what is actually there. Taken together, these results show that CountBCF can recover established radiation-epidemiology structure, quantify its uncertainty, and expose interpretable departures for further investigation.

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